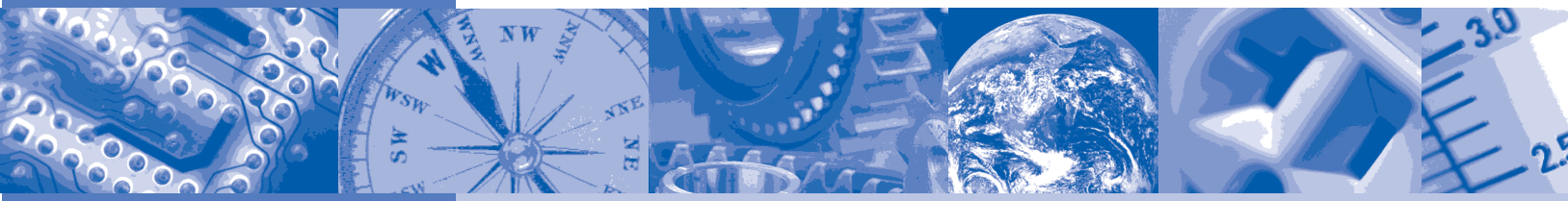
TECHNICAL REPORT

ISA–TR20.00.01–2001



Specification Forms for Process Measurement and Control Instruments Part 1: General Considerations

Approved 04 April 2001

ISA–The Instrumentation, Systems, and Automation Society

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ISA–TR20.00.01–2001
Specification Forms for Process Measurement and Control Instruments
Part 1: General Considerations

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Preface

This preface, as well as all footnotes and annexes, is included for information purposes and is not part of ISA–TR20.00.01–2001.

The standards referenced within this document may contain provisions that, through reference in this text, constitute requirements of this document. At the time of publication, the editions indicated were valid. All standards are subject to revision, and parties to agreements based on this document are encouraged to investigate the possibility of applying the most recent editions of the standards indicated within this document. Members of IEC and ISO maintain registers of currently valid International Standards. ANSI maintain registers of currently valid U.S. National Standards.

This document has been prepared as part of the service of ISA–The Instrumentation, Systems, and Automation Society, toward a goal of uniformity in the field of instrumentation. To be of real value, this document should not be static but should be subject to periodic review. Toward this end, the Society welcomes all comments and criticisms and asks that they be addressed to the Secretary, Standards and Practices Board; ISA; 67 Alexander Drive; P. O. Box 12277; Research Triangle Park, NC 27709; Telephone (919) 549-8411; Fax (919) 549-8288; E-mail: standards@isa.org.

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Foreword

This technical report has been prepared by ISA's SP20 standards committee, Instrument Specification Forms. The ISA-SP20 committee welcomes comments and suggestions, and requests that they be addressed to the Chairperson, ISA-SP20, ISA, 67 Alexander Drive, Research Triangle Park, NC 27709 USA.

This technical report provides separate form parts for operating parameters, device specifications, and general requirements. Forms from ISA-20-1981 have been modified extensively, and many of the forms represent new devices not previously covered.

This technical report applies to all processes of development and use of ISA specification forms for process measurement and control instruments. It provides the listing of the forms, the classification of the devices, and the approved forms.

The other parts of the series, *Specification Forms for Process Measurement and Control Instruments,* are as follows:

ISA Draft 20.00.02	Part 2: Instructions for Using Specification Forms
ISA–20.00.03–2001	Part 3: Form Requirements and Development Guidelines
ISA Draft TR20.00.04	Part 4: Data Dictionary

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Introduction

The completion of instrument specification forms is an extremely important segment of the complete design, purchase, and manufacture of process measurement and control instrumentation.

Approved ISA specification forms for many devices are developed that follow the guidelines of ISA–20.00.03–2001 and have been reviewed by users and manufacturers. These forms are adequate for most applications and can be provided by ISA. The defining of specification forms and a data dictionary will allow electronic data exchange between users and manufacturers.

The purposes of ISA specification forms are

a) to assist in preparation of a complete specification by listing and providing space for all principal descriptive attributes;

b) to facilitate quoting, bid reviews, purchasing, receiving, calibration, inspection, piping design, design audit, accounting, and ordering procedures by uniform display of information;

c) to improve efficiency of instrumentation activities from initial concept to final commissioning and any subsequent reviews and revisions; and

d) to provide standard data definitions and field lengths adequate for electronic exchange of information.

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1 Scope

Instrument devices with dedicated functionality may be specified using data-sheet-type, approved ISA specification forms. This technical report does not address system specifications such as shared control and shared display devices, where extensive use of functionality or performance specifications is required.

This technical report lists the ISA specification forms by form part category, measured or initiating variable, readout, or final control device categories, and form number. A form numbering system has been implemented that identifies the form categories, device function grouping, and page number. The form number, title, and development status is listed for each proposed or approved form.

2 Normative references

ISA–20.00.03–2001, Specification Forms for Process Measurement and Control Instruments, Part 3: Form Requirements and Development Guidelines.

3 Definitions

For the purpose of this technical report, the following definition applies:

3.1 specification form:
the document, or its electronic data equivalent, that details the statement of parameters or properties prescribing the design basis, construction, materials, and performance for an instrument to be supplied.

4 Application

ISA specification forms may be used for the following different levels of device specification.

4.1 Operating parameters documentation

Documentation of the operating parameters may be completed and endorsed independently of the hardware device selection and preliminary specification through the use of the separate form page.

4.2 Preliminary inquiry/quotation

For a preliminary inquiry or quotation, the specifier and vendor may agree on a minimum level of information. In this case, the specifier may mark with an asterisk (*) in the revision column of the *Operating Parameters* or *Device Specification* forms, any lines that the recipient of the specification form is requested to complete.

4.3 Traditional specification

Documentation of operating parameters required for sizing or selection analysis, and identification of the instrument properties that are pertinent to the development of a manufacturer's model number, should be the minimum data provided on specification forms used to purchase instruments.

4.4 Conforming specification

A complete specification ideally should include all relevant information about the main device and associated secondary devices that has been agreed to by the purchaser and supplier. In this case, the fully completed specification form can be used for the final documentation of the instrument specifications.

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5 Requirements

5.1 Three form parts

ISA specification forms are generally comprised of three form parts for the following information organization:

a) *Operating Parameters* forms are developed to document the process application and design conditions for communication to instrument specifiers and manufacturers.

b) *Device Specification* forms are developed to provide the information to initially size and select a main device and the information to identify requirements for attached and directly associated devices.

c) *General or Special Requirements* forms are developed to provide means for the specifier to identify general or special requirements that are applicable to the device, but need additional space beyond that available in the other two types of forms.

5.2 Form numbers and titles for *Operating Parameters* forms

A common form is developed for all devices sharing the same measured variable or final control element. Table A.1, annex A, identifies assigned categories, ISA form numbers, and titles.

5.3 Form numbers and titles for *Device Specification* forms

Device Specification forms are grouped and numbered based upon a device category code, the form part number, and an instrument function subcategory sequential number. Table A.2, annex A, identifies assigned device categories, ISA form numbers, and titles.

5.4 Form numbers and titles for *General or Special Requirements* forms

A single form is provided for the initial page of the *General or Special Requirements* forms. Table A.3, annex A, identifies the assigned ISA form number and title.

6 General sections of ISA specification forms

6.1 Specification identification section

Operating Parameters, Device Specification, and *General or Special Requirements* forms utilize the same specification identification field prompts.

6.2 Endorsements

Endorsement identifications and status inference, such as originator, checker, approver, and origin date, should be independent throughout each form type.

6.3 Revision chronicle

Identification of the three most recent entries recorded in the revision chronicle should be independent throughout each form type.

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6.4 Component identification

Identification of the instrument, such as primary tag number, description, line/equipment number, and reference drawing number, should be identical throughout each form type.

7 Classification and designation

Hardware specification forms are based on specific hardware types as classified in annex A.

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Annex A (informative) – Specification forms tables of contents

Table A.1 – Operating parameter device categories and form titles

Form Number	Form Title	Development Status, ISA-SP20
20A1001	Analysis Device	In process
20A1002	Analysis Device Components	In process
20B1001	Burner or Combustion Device	
20C1001	Valve or Regulator Device	Approved (Annex B)
20F1001	Flow Device	Approved (Annex B)
20H1001	Hand Control Device	
20L1001	Level Device	Approved (Annex B)
20P1001	Pressure or Differential Pressure Device	Approved (Annex B)
20P1002	Pressure Safety Device	In process
20R1001	Receiver Device	
20T1001	Temperature Device	Approved (Annex B)
20U1001	Basic Engineering Design Data	In process
20V1001	Vibration or Mechanical Analysis Device	
20W1001	Weight Device	
20X1001	Unclassified Device	
20Y1001	Relay or Compute or Converter Device	
20Z1001	Position Device	

Table A.2 – Device specification categories and form titles

Form Number	Form Title	Development Status, ISA-SP20
Analysis Devices		
20A2101	Carbon Dioxide Monitor	
20A2111	Combustible Gas Analyzer	In process
20A2121	Flue & Stack Gas Analyzer	
20A2141	Chromatograph Analyzer	In process
20A2142	Chromatograph Analyzer Components	In process
20A2151	In Situ Oxygen Analyzer	In process
20A2161	Extractive Oxygen Analyzer	In process
20A2162	Extractive Oxygen Analyzer Components	In process
20A2171	Opacity Analyzer	
20A2181	Toxic Gas Analyzer	

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Table A.2 – Device specification categories and form titles(cont'd)

Analysis Devices (cont'd)		
20A2191	Dissolved Oxygen Analyzer	
20A2201	Infrared or Ultraviolet Analyzer	In process
20A2202	Infrared or Ultraviolet Analyzer Components	In process
20A2301	Ion-Selective Analyzer	
20A2311	Liquid Density or Specific Gravity Transmitter	In process
20A2321	Nuclear-Type Density Transmitter	In process
20A2331	Oil-In-Water Detector	
20A2341	PH or ORP Transmitter w/wo Insertion Assembly	
20A2351	Turbidity or Colorimeter Analyzer	In process
20A2361	Viscometer	In process
20A2371	Moisture Analyzer	In process
20A2372	Moisture Analyzer Components	
20A2381	Conductivity Analyzer w/wo Insertion Assembly	In process
20A2901	Sample Conditioning System w/wo Switches and Valves	In process
Burner or Combustion Devices		
20B2001	Flame Detector	
20B2101	Flame Monitor	
Valve or Regulator Devices		
20C2001	Linear Motion Type Control Valve Assembly	Approved (Annex B)
20C2002	Linear Motion Type Control Valve Auxiliaries	Approved (Annex B)
20C2011	Rotary Motion Type Control Valve Assembly	Approved (Annex B)
20C2012	Rotary Motion Type Control Valve Auxiliaries	Approved (Annex B)
20C2021	Severe Service Control Valve Assembly	In process
20C2022	Severe Service Control Valve Auxiliaries	In process
20C2031	Desuperheater Control Valve	
20C2041	Motor-Operated Valve Assembly	In process
20C2042	Motor-Operated Valve Auxiliaries	In process
20C2101	Recirculation Control Valve	
20C2111	Pressure Regulator w/wo Pilot Valve	In process
20C2121	Pressure Relief Valve w/wo Pilot Valve	In process
20C2131	Temperature Regulator Valve	In process
20C2201	Power-Actuated Block Valve Assembly	In process
20C2202	Power-Actuated Block Valve Auxiliaries	In process
20C2211	Solenoid Valve	
20C2221	Manual-Actuated Block Valve Assembly	

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Table A.2 – Device specification categories and form titles (cont'd)

Flow Devices		
20F2001	Anemometer Element	
20F2011	Elbow Meter	
20F2021	Flow Nozzle w/wo Meter Tube	Approved (Annex B)
20F2031	Orifice Plate	Approved (Annex B)
20F2041	Orifice Plate Assembly w/wo Meter Tube	Approved (Annex B)
20F2051	Multivariable Flow Transmitter	Approved (Annex B)
20F2061	Paddle Flow Switch	
20F2071	Pitot Tube w/wo Insertion Assembly	In process
20F2081	Segmental Wedge Flowmeter	
20F2091	Target Flowmeter	
20F2101	V-Cone Flow Element	
20F2111	Venturi or Flow Tube w/wo Meter Tube	Approved (Annex B)
20F2211	Flume Flowmeter	
20F2221	Open Channel Flowmeter	
20F2231	Rotameter Flowmeter	In process
20F2241	Rotameter Transmitter	
20F2251	Variable Area Flowmeter	
20F2261	Weir Flowmeter	
20F2311	Doppler Flowmeter	
20F2321	Magnetic Flowmeter w/wo Integral Totalizer Indicator	Approved (Annex B)
20F2331	Nuclear Type Flowmeter	
20F2341	Sonic or Ultrasonic Flowmeter	In process
20F2351	Thermal Flowmeter	
20F2361	Thermal Dispersion Flow Switch	
20F2371	Turbine Flowmeter w/wo Totalizer Indicator	In process
20F2381	Vortex or Swirl Flowmeter w/wo Totalizer Indicator	Approved (Annex B)
20F2411	Positive Displacement Flowmeter Direct-Reading Type	In process
20F2421	Positive Displacement Flow Transmitter	In process
20F2431	Variable Area Flow Switch	In process
20F2511	Mass Type Flowmeter	
20F2521	Coriolis Mass Flowmeter w/wo Totalizer Indicator	Approved (Annex B)
20F2611	Sight Flow Glass	In process
Hand Control Devices		
20H2001	Instrument Valve	In process
20H2011	Instrument Manifold Valve	

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Table A.2 – Device specification categories and form titles (cont'd)

Level Devices		
20L2101	Capacitance or RF Admittance Level Transmitter w/wo Switches	Approved (Annex B)
20L2111	Differential Pressure Level Transmitter - Flange Mounted	Approved (Annex B)
20L2121	Displacer-Type Level Transmitter or Local Controller	Approved (Annex B)
20L2131	Tape Level Gauge or Transmitter w/wo Switches	In process
20L2141	Nuclear Radiation Level Transmitter w/wo Switches	Approved (Annex B)
20L2151	Resistance-Tape Level Transmitter w/wo Switches	Approved (Annex B)
20L2161	Radar Level Transmitter w/wo Switches	
20L2171	Non-contact Ultrasonic Level Transmitter w/wo Switches	Approved (Annex B)
20L2181	Magnetostrictive Level Transmitter w/wo Switches	In process
20L2301	Capacitance or RF Admittance Level Switch	Approved (Annex B)
20L2311	Float or Displacer Level Switch	Approved (Annex B)
20L2312	Tank Level Gauge or Indicator w/wo Switches	Approved (Annex B)
20L2321	Nuclear Radiation Level Switch	Approved (Annex B)
20L2331	Resistance Tape Level Switch	
20L2341	RF or Radar or Micro Level Switch	
20L2351	Ultrasonic Contact-Type Level Switch	Approved (Annex B)
20L2361	Vibrating Element Level Switch	
20L2371	Rotary Level Switch	Approved (Annex B)
20L2501	Liquid Level Gage Glass w/wo Illuminator	Approved (Annex B)
20L2511	Magnetic Liquid Level Gage or Indicator w/wo Switches	Approved (Annex B)
20L2521	Tape Float Tank Gauge	
Pressure or Differential Pressure Devices		
20P2001	Pressure Gauge	Approved (Annex B)
20P2011	Pressure Gauge With Diaphragm Seal	Approved (Annex B)
20P2101	Differential Pressure Gauge	Approved (Annex B)
20P2201	Pressure Transmitter	Approved (Annex B)
20P2211	Pressure Transmitter With Diaphragm Seal	Approved (Annex B)
20P2301	Differential Pressure Transmitter	Approved (Annex B)
20P2311	Differential Pressure Transmitter With Diaphragm Seals	Approved (Annex B)
20P2401	Pressure Switch w/wo Transmitter	Approved (Annex B)
20P2501	Differential Pressure Switch w/wo Transmitter	Approved (Annex B)
Pressure Safety Devices		
20P2901	Rupture Disk	In process
20P2911	Pressure Relief Tank Vent	
20P2921	Pressure Safety or Relief Valve	In process

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Table A.2 – Device specification categories and form titles (cont'd)

Receiver Devices		
20R2011	Annunciator	
20R2111	Controller	
20R2121	Local Controller w/wo Recorder	
20R2201	Receiver Gauge	Approved (Annex B)
20R2211	Indicator	
20R2221	Local Indicator w/wo Recorder	
20R2231	Totalizer Indicator	
20R2241	Digital Meter	
20R2311	Analog Recorder	
20R2411	Digital Recorder	
Temperature Devices		
20T2001	Bimetallic Thermometer w/wo Thermowell	Approved (Annex B)
20T2101	Filled-System Thermometer Assembly w/wo Thermowell	Approved (Annex B)
20T2111	Filled-System Temperature Switch w/wo Thermowell	Approved (Annex B)
20T2121	Filled-System Temperature Transmitter w/wo Thermowell	Approved (Annex B)
20T2201	RTD Assembly w/wo Thermowell	Approved (Annex B)
20T2221	RTD/Thermocouple Temperature Transmitter or Switch	Approved (Annex B)
20T2301	Thermocouple Assembly w/wo Thermowell	Approved (Annex B)
20T2401	Thermal Radiation Temperature Sensor w/wo Monitor	Approved (Annex B)
20T2501	Thermowell or Protecting Tube Assembly	Approved (Annex B)
Vibration or Mechanical Analysis Devices		
20V2111	Vibration Switch	
20V2211	Vibration Monitor	
20V2311	Mechanical Analysis Monitor	
Weight Devices		
20W2011	Electronic Weight Scale	
20W2111	Strain Gage Weight Indicator	
20W2211	Load Cell Weight Transmitter	
20W2221	Strain Gage Weight Transmitter	
Relay or Compute or Converter Devices		
20Y2011	Transducer	
20Y2111	Barrier or Isolator	
20Y2211	Signal Converter	
20Y2311	Analog Signal Switch	
20Y2411	Solenoid or Pilot Valve	

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Table A.2 – Device specification categories and form titles (cont'd)

Position Devices		
20Z2011	Mechanical Limit Switch	
20Z2021	Proximity Switch	
20Z2111	Position Transmitter	

Table A.3- General or special requirements form title

Form Number	Form Title	Development Status, ISA-SP20
20G3001	General or Special Requirements	Approved (Annex B)

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Annex B (informative) – Approved specification forms

Following are the approved specification forms, as of the date of this publication.

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RESPONSIBLE ORGANIZATION	VALVE OR REGULATOR DEVICE	SPECIFICATION IDENTIFICATIONS
	Operating Parameters	Document no
		Latest revision / Date
		Issue status

ADMINISTRATIVE IDENTIFICATIONS			
Project number		Sub project no	
Project			
Enterprise			
Site			
Area		Cell / Unit	

SERVICE IDENTIFICATIONS			
Tag no/Functional ident			
Related equipment			
Service			
P&ID/Reference dwg number			
Upstr line/nozzle number			
Upstream line pipe spec			
Upstr line nominal size		Rating	
Upstr line conn type		Style	
Upstr line schedule no		Wall thickness	
Upstr conn orientation			
Upstr line material type			
Connection design code			
Dnstr line/nozzle number			
Downstream line pipe spec			
Dnstr line nominal size		Rating	
Dnstr line conn type		Style	
Dnstr line schedule no		Wall thickness	
Dnstr conn orientation			
Dnstr line material type			
Dnstr insulation atten			

SERVICE IDENTIFICATIONS Continued			
Inline hazardous area cl		Div/Zone	Group
Inline area min ign temp		Temp ident number	
Remote hazardous area cl		Div/Zone	Group
Remote area min ign temp		Temp ident number	

COMPONENT DESIGN CRITERIA			
Component type			
Component style			
Input signal type			
Flow characteristic			
Min seat leakage class			
Max stem leakage rate			
Type of protection			
Criticality code			
Max EMI susceptibility		Ref	
Min xdcr reqd accuracy		Ref	
Max offset/propn band			
Actuator sizing factor			
Avail nom power supply		Number wires	
Available nom air supply			
Testing/Listing agency			
Test requirements			
Supply loss failure mode			
Signal loss failure mode			

PROCESS VARIABLES	MATERIAL FLOW CONDITIONS				PROCESS DESIGN CONDITIONS		
Flow Case Identification				Units	Minimum	Maximum	Units
Inlet pressure							
Outlet pressure							
Inlet temperature							
Inlet phase type							
Mass fraction vapor							
Total mass flow rate							
Liquid mass flow rate							
Liquid actual flow rate							
Liquid standard flow rate							
Liquid density							
Liquid specific gravity							
Liquid viscosity							
Absolute vapor pressure							
Vapor mass flow rate							
Vapor actual flow rate							
Vapor standard flow rate							
Vapor density							
Vapor specific gravity							
Vapor molecular weight							
Vapor viscosity							
Inlet compressibility							
CALCULATED VARIABLES							
Pressure differential							
Total system press drop							
Calculated flow coef							
Calculated travel							
Calculated travel time							
Calculated SPL							

MATERIAL PROPERTIES			
Name			
Composition			
Density at ref temp		At	
Ratio sp heat capacity			

MATERIAL PROPERTIES Continued			
Abs critical pressure			
Critical temperature			
NFPA health hazard		Flammability	Reactivity

Rev	Date	Revision Description	By	Appv1	Appv2	Appv3	REMARKS

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1		RESPONSIBLE ORGANIZATION	FLOW DEVICE		6		SPECIFICATION IDENTIFICATIONS
2					7		Document no
3			Operating Parameters		8		Latest revision / Date
4					9		Issue status
5					10		

11		ADMINISTRATIVE IDENTIFICATIONS		40		SERVICE IDENTIFICATIONS Continued	
12		Project number	Sub project no	41		Inline hazardous area cl	Div/Zone / Group
13		Project		42		Inline area min ign temp	Temp ident number
14		Enterprise		43		Remote hazardous area cl	Div/Zone / Group
15		Site		44		Remote area min ign temp	Temp ident number
16		Area	Cell / Unit	45			
17				46			
18		SERVICE IDENTIFICATIONS		47		COMPONENT DESIGN CRITERIA	
19		Tag no/Functional ident		48		Component type	
20		Related equipment		49		Component style	
21		Service		50		Output signal type	
22				51		Characteristic curve	
23		P&ID/Reference dwg number		52		Compensation style	
24		Upstr line/nozzle number		53		Type of protection	
25		Upstream line pipe spec		54		Criticality code	
26		Upstr line nominal size	Rating	55		Max EMI susceptibility	Ref
27		Upstr line conn type	Style	56		Max temperature effect	
28		Upstr line schedule no	Wall thickness	57		Min diameter ratio (d/D)	Max
29		Upstr conn orientation		58		Max response time	
30		Upstr line material type		59		Min required accuracy	Ref
31		Connection design code		60		Avail nom power supply	Number wires
32		Dnstr line/nozzle number		61		Min load capability	
33		Downstream line pipe spec		62		Testing/Listing agency	
34		Dnstr line nominal size	Rating	63		Test requirements	
35		Dnstr line conn type	Style	64		Supply loss failure mode	
36		Dnstr line schedule no	Wall thickness	65		Signal loss failure mode	
37		Dnstr conn orientation		66			
38		Dnstr line material type		67			
39		Avail upstr straight lg	Dnstr lg	68			

69		PROCESS VARIABLES	MATERIAL FLOW CONDITIONS				101		PROCESS DESIGN CONDITIONS
70		Flow Case Identification				Units	102		Minimum / Maximum / Units
71		Inlet pressure					103		
72		Outlet pressure					104		
73		Inlet temperature					105		
74		Inlet phase type					106		
75		Mass fraction vapor					107		
76		Total mass flow rate					108		
77		Liquid mass flow rate					109		
78		Liquid actual flow rate					110		
79		Liquid standard flow rate					111		
80		Liquid density					112		
81		Liquid specific gravity					113		
82		Liquid viscosity					114		
83		Absolute vapor pressure					115		
84		Vapor mass flow rate					116		
85		Vapor actual flow rate					117		
86		Vapor standard flow rate					118		
87		Vapor density					119		
88		Vapor specific gravity					120		
89		Vapor molecular weight					121		
90		Vapor viscosity					122		
91		Inlet compressibility					123		
92							124		
93		CALCULATED VARIABLES					125		
94		Pressure differential					126		
95		Perm pressure drop					127		
96		Line fluid velocity					128		
97		Line Reynolds number					129		
98		Calculated uncertainty					130		
99							131		

132		MATERIAL PROPERTIES		138		MATERIAL PROPERTIES Continued	
133		Name		139		Abs critical pressure	
134		Composition		140		Critical temperature	
135		Density at ref temp	At	141		NFPA health hazard	Flammability / Reactivity
136		Ratio sp heat capacity		142			
137		Conductivity		143			

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2				7		Document no	
3			Operating Parameters	8		Latest revision	Date
4				9		Issue status	
5				10			

11		ADMINISTRATIVE IDENTIFICATIONS		43		SERVICE IDENTIFICATIONS Continued	
12		Project number	Sub project no	44		Process conn diff height	
13		Project		45		Zero datum height	
14		Enterprise		46		Turbulence intensity	
15		Site		47		Inline hazardous area cl	Div/Zone / Group
16		Area	Cell / Unit	48		Inline area min ign temp	Temp ident number
17				49		Remote hazardous area cl	Div/Zone / Group
18		SERVICE IDENTIFICATIONS		50		Remote area min ign temp	Temp ident number
19		Tag no/Functional ident		51			
20		Related equipment		52			
21		Service		53			
22				54		COMPONENT DESIGN CRITERIA	
23		P&ID/Reference dwg number		55		Component type	
24		Upper conn nozzle number		56		Component style	
25		Upper conn pipe spec		57		Output signal type	
26		Upper conn nominal size	Rating	58		Characteristic curve	
27		Upper conn termn type	Style	59		Compensation required	
28		Upper conn schedule no	Wall thickness	60		Type of protection	
29		Upper conn length		61		Criticality code	
30		Upper conn orientation		62		Max EMI susceptibility	Ref
31		Upper conn material type		63		Max temperature effect	Ref
32		Connection design code		64		Max response time	
33		Lower conn nozzle number		65		Min required accuracy	Ref
34		Lower conn pipe spec		66		Max dead band	
35		Lower conn nominal size	Rating	67		Avail nom power supply	Number wires
36		Lower conn termn type	Style	68		Min load capability	
37		Lower conn schedule no	Wall thickness	69		Testing/Listing agency	
38		Lower conn length		70		Test requirements	
39		Lower conn orientation		71		Supply loss failure mode	
40		Lower conn material type		72		Signal loss failure mode	
41		Device insertion length		73			
42		Vessel nominal diameter	Height	74			

75		PROCESS VARIABLES	PROCESS DESIGN CONDITIONS				
76		Condition Identification	Minimum	Normal	Maximum	Units	
77		Pressure					
78		Temperature					
79		Lower material phase type					
80		Lower material density					
81		Lower material sp gr					
82		Lower material viscosity					
83		Lower material level					
84		Upper material phase type					
85		Upper material density					
86		Upper material sp gr					
87		Upper material viscosity					
88		Upper material level					
89		Max material buildup thkns					
90		Solids concentration					
91		Moisture content					
92		Max angle of repose					
93		Particle size					
94							
95							
96		CALCULATED VARIABLES					
97		Calc zero el/suppr					
98		Calc measurement span					
99							
100							

101		MATERIAL PROPERTIES		109		MATERIAL PROPERTIES Continued	
102		Lower material name		110		Upper material name	
103		Lower matl composition		111		Upper matl composition	
104		Lower matl density at ref	At	112		Upper matl density at ref	At
105		Lower dielectric constant	At temp	113		Upper dielectric constant	At temp
106		Lower material Color		114		Upper material Color	
107		Lower NFPA health hazard	Flammability / Reactivity	115		Upper NFPA health hazard	Flammability / Reactivity
108				116			

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		Document no
		Latest revision / Date
		Issue status

ADMINISTRATIVE IDENTIFICATIONS	
Project number	Sub project no
Project	
Enterprise	
Site	
Area	Cell / Unit

SERVICE IDENTIFICATIONS	
Tag no/Functional ident	
Related equipment	
Service	
P&ID/Reference dwg number	
High side line/nozzle no	
High side conn pipe spec	
High side conn nom size	Rating
High side conn termn type	Style
High side conn schedule	Wall thickness
High side conn length	
High side conn orient	
High side conn matl type	
Connection design code	
Low side line/nozzle no	
Low side conn pipe spec	
Low side conn nom size	Rating
Low side conn termn type	Style
Low side conn schedule no	Wall thickness
Low side conn length	
Low side conn orientation	
Low side conn matl type	
Device insertion length	

SERVICE IDENTIFICATIONS Continued	
Process conn diff length	
Zero datum conn length	
Inline hazardous area cl	Div/Zone / Group
Inline area min ign temp	Temp ident number
Remote hazardous area cl	Div/Zone / Group
Remote area min ign temp	Temp ident number

COMPONENT DESIGN CRITERIA	
Component type	
Component style	
Output signal type	
Characteristic curve	
Type of protection	
Criticality code	
Max EMI susceptibility	
Max temperature effect	
Max response time	
Min required accuracy	Ref
Max dead band	
Avail nom power supply	Number wires
Min load capability	
Testing/Listing agency	
Test requirements	
Supply loss failure mode	
Signal loss failure mode	

PROCESS VARIABLES	PROCESS DESIGN CONDITIONS			
Condition Identification	Minimum	Normal	Maximum	Units
High side pressure				
Low side pressure				
High side temperature				
Low side temperature				
High side matl phase type				
High side material viscosity				
Low side matl phase type				
Low side material viscosity				
Differential pressure				
CALCULATED VARIABLES				
Calc zero el / suppr				
Calc measurement span				

MATERIAL PROPERTIES	
Low side material name	
Low side composition	
Low side density at ref	At
Low side NFPA health haz	Flammability / Reactivity

MATERIAL PROPERTIES Continued	
High side material name	
High side composition	
High side density at ref	At
High side NFPA health haz	Flammability / Reactivity

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RESPONSIBLE ORGANIZATION	TEMPERATURE DEVICE	SPECIFICATION IDENTIFICATIONS
	Operating Parameters	Document no
		Latest revision / Date
		Issue status

ADMINISTRATIVE IDENTIFICATIONS			
Project number		Sub project no	
Project			
Enterprise			
Site			
Area		Cell / Unit	

SERVICE IDENTIFICATIONS			
Tag no/Functional ident			
Related equipment			
Service			
P&ID/Reference dwg number			
Line/Nozzle number			
Process conn pipe spec			
Line/Nozzle nominal size		Rating	
Process conn termn type		Style	
Line/Nozzle schedule no		Wall thickness	
Connection length			
Connection orientation			
Connection material type			
Connection design code			
Insulation thickness			
Device insertion length			

SERVICE IDENTIFICATIONS Continued			
Inline hazardous area cl		Div/Zone	Group
Inline area min ign temp		Temp ident number	
Remote hazardous area cl		Div/Zone	Group
Remote area min ign temp		Temp ident number	

COMPONENT DESIGN CRITERIA			
Component type			
Component style			
Output signal type			
Characteristic curve			
Type of protection			
Criticality code			
Max EMI susceptibility		Ref	
Max temperature effect		Ref	
Max response time			
Min required accuracy		Ref	
Max dead band			
Avail nom power supply		Number wires	
Minimum load capability			
Testing/Listing agency			
Test requirements			
Supply loss failure mode			
Signal loss failure mode			

PROCESS VARIABLES	MATERIAL FLOW CONDITIONS				PROCESS DESIGN CONDITIONS		
Flow Case Identification				Units	Minimum	Maximum	Units
Pressure							
Temperature							
Phase type							
Total mass flow rate							
Liquid mass flow rate							
Liquid actual flow rate							
Liquid standard flow rate							
Liquid density							
Liquid specific gravity							
Liquid viscosity							
Vapor mass flow rate							
Vapor actual flow rate							
Vapor standard flow rate							
Vapor density							
Vapor specific gravity							
Vapor molecular weight							
CALCULATED VARIABLES							
Line fluid velocity							
Line Reynolds number							
Wake/natural freq ratio							

MATERIAL PROPERTIES			
Name			
Composition			
Density at ref temp		At	
Ratio sp heat capacity			

MATERIAL PROPERTIES Continued			
Ratio sp heat factor			
Emissivity			
NFPA health hazard		Flammability	Reactivity

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RESPONSIBLE ORGANIZATION	LINEAR MOTION TYPE CONTROL VALVE ASSEMBLY Device Specification	SPECIFICATION IDENTIFICATIONS
		Document no
		Latest revision / Date
		Issue status

VALVE BODY AND BONNET			
Body type			
Bonnet style			
Inlet conn nominal size		Rating	
Outlet conn nominal size		Rating	
End conn termn type		Style	
Flange facing finish			
Stem seal type			
Face-to-face standard			
Yoke boss diameter			
Body and bonnet material			
Lining material			
Body/Bonnet bolting matl			
Packing flg bolting matl			
Gasket material			
Packing material			
Bellows seal material			

VALVE TRIM			
Closure member type			
Trim style			
Inherent characteristic			
Seat ring type			
Flow direction force			
Port/Orifice diameter			
Stem diameter		Rt travel	
Closure member material			
Seat ring material			
Stem material			
Cage material			
Guide/Retainer material			
Guide bushing material			
Hard facing/coating			

ACTUATOR			
Actuator type			
Acting style			
Effective area			
Bench set LRL		URL	
Max supply capability			
Handwheel type/position			
Casing/cylinder material			
Diaphragm/O ring matl			
Yoke material			

SIGNAL TRANSDUCER	
Input signal type	
Output signal type	
Accuracy rating	

SIGNAL TRANSDUCER continued	
Enclosure type no/class	
Digital communication std	
Signal power source	
Pressure gauges	
Cert/Approval type	
Mounting location/type	
Enclosure material	

POSITIONER OR CONTROLLER	
Configuration type	
Input signal type	
Enclosure type no/class	
Signal power source	
Digital communication std	
Cert/Approval type	
Pressure gauges	
By-pass manifold	
Enclosure material	

PERFORMANCE CHARACTERISTICS			
Max press at design temp		At	
Min working temperature		Max	
Max pressure drop			
Rated flow coef type	Coef		Xt
Rated factor Fl		Sigma mr	
Rated seat leakage class			

ACCESSORIES	
Air set filter style	
Air set gauges	
Tubing nominal size	
Tubing and fitting matl	

SPECIAL REQUIREMENTS	
Custom tag	
Reference specification	
Special preparation	
Compliance standard	
Construction code	
Software configuration	

PHYSICAL DATA	
Estimated weight	
Face-to-face dimension	
Overall height	
Removal clearance	
Mfr reference dwg	

CALIBRATIONS AND TEST		INPUT OR TEST			OUTPUT	
TAG NO/FUNCTIONAL IDENT	MEAS/SIGNAL/TEST	LRV	URV	ACTION	LRV	URV
	Bench set-Travel					
	Positioner input-Travel					
	Transducer input-Output					
	Test pressure					

COMPONENT IDENTIFICATIONS		
COMPONENT TYPE	MANUFACTURER	MODEL NUMBER

Rev	Date	Revision Description	By	Appv1	Appv2	Appv3	REMARKS

Form: 20C2001 Rev 0

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RESPONSIBLE ORGANIZATION	LINEAR MOTION TYPE CONTROL VALVE AUXILIARIES Device Specification	SPECIFICATION IDENTIFICATIONS
		Document no
		Latest revision Date:
		Issue status

POSITION SWITCHES			
Housing type			
Sensor style			
Output signal type			
Enclosure type no/class			
Contacts ac rating		At Max	
Contacts dc rating		At Max	
Contacts arrangement		Quantity	
Integral indicator style			
Signal termination type			
Cert/Approval type			
Mounting location/type			
Enclosure material			
Shaft material			
Bushing material			
Fastener material			
Gasket/O ring material			

SOLENOID OR PILOT VALVE			
Body type			
Enclosure style			
Inlet conn nominal size		Style	
Coil watt rating			
Coil insulation class			
Enclosure type no/class			
Min operating press diff		Max	
Signal power source			
Manual operator type			
Type of actuation			
Coil type			
Operation style			
Press port orifice size		Exhaust port	
Signal termination type			
Cert/Approval type			
Mounting location/type			
Enclosure material			
Body material			
Seat/Disc/Diaphragm matl			

POSITION TRANSMITTER			
Housing type			
Sensor style			
Output signal type			
Accuracy/Linearity rating			
Enclosure type no/class			
Digital communication std			
Signal power source			

POSITION TRANSMITTER continued			
Integral indicator style			
Signal termination type			
Cert/Approval type			
Mounting location/type			
Enclosure material			

SUPPLY PRESSURE TRIP VALVE			
Configuration type			
Volume tank size			
Trip set pressure reset			
Tank max working pressure			
Min supply capability		Max	
Trip set pressure LRL		URL	
Construction code			
Mounting location/type			
Body material			
Diaphragm/O ring material			

EMISSIONS MONITOR SWITCH			
Sensor type			
Housing type			
Inlet conn nominal size		Style	
Output signal type			
Enclosure type no/class			
Contacts ac rating		At Max	
Contacts dc rating		At Max	
Contacts arrangement			
Integral indicator style			
Signal termination type			
Cert/Approval type			
Mounting location/type			
Enclosure material			
Diaphragm material			

BOOSTER RELAY			
Configuration type			
End conn nominal size		Style	
Dead band rating			
Max supply capability			
Supply port size		Exh port	
Body material			
Closure member matl			
Diaphragm material			
Test conn material			
Gasket/O ring material			

CALIBRATIONS AND TEST		INPUT OR SETPOINT			OUTPUT	
TAG NO/FUNCTIONAL IDENT	MEAS/SIGNAL/TEST	LRV	URV	ACTION	LRV	URV
	Position-Analog output					
	Solenoid energized-Output					
	Position setpoint-Output					
	Supply setpoint–Output					
	Emissions setpoint–Output					

COMPONENT IDENTIFICATIONS		
COMPONENT TYPE	MANUFACTURER	MODEL NUMBER
Position transmitter		
Solenoid/Pilot valve		
Position switch		
Supply press trip valve		
Emissions monitor		
Booster relay		

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		Document no	
		Latest revision	Date
		Issue status	

VALVE BODY AND BONNET			
Body type			
Bonnet style			
Inlet conn nominal size		Rating	
Outlet conn nominal size		Rating	
End conn termn type		Style	
Flange facing finish			
Shaft seal type			
Face-to-face standard			
Yoke boss diameter			
Body and bonnet material			
Lining material			
Body/Bonnet bolting matl			
Packing flg bolting matl			
Gasket material			
Packing material			

VALVE TRIM			
Closure member type			
Trim style			
Inherent characteristic			
Seal ring type			
Flow direction force			
Shaft diameter		Rated travel	
Closure member material			
Seal ring material			
Shaft material			
Flow ring material			
Guide/Retainer material			
Guide bushing material			
Hard facing/coating			

ACTUATOR			
Actuator type			
Effective area			
Bench set LRL		URL	
Max supply capability			
Handwheel type/position			
Mounting orientation			
Casing/cylinder material			
Diaphragm/O ring matl			
Yoke material			

SIGNAL TRANSDUCER	
Input signal type	
Output signal type	
Accuracy rating	

SIGNAL TRANSDUCER continued	
Enclosure type no/class	
Digital communication std	
Signal power source	
Pressure gauges	
Cert/Approval type	
Mounting location/type	
Enclosure material	

POSITIONER OR CONTROLLER	
Configuration type	
Input signal type	
Enclosure type no/class	
Signal power source	
Digital communication std	
Cert/Approval type	
Pressure gauges	
By-pass manifold	
Enclosure material	

PERFORMANCE CHARACTERISTICS				
Max press at design temp		At		
Min working temperature		Max		
Max pressure drop				
Rated flow coef type		Coef		Xt
Rated factor Fl		Sigma mr		
Rated seat leakage class				

ACCESSORIES	
Air set filter style	
Air set gauges	
Tubing nominal size	
Tubing and fitting matl	

SPECIAL REQUIREMENTS	
Custom tag	
Reference specification	
Special preparation	
Compliance standard	
Construction code	
Software configuration	

PHYSICAL DATA	
Estimated weight	
Face-to-Face dimension	
Overall height	
Removal clearance	
Mfr reference dwg	

CALIBRATIONS AND TEST		INPUT OR TEST			OUTPUT	
TAG NO/FUNCTIONAL IDENT	MEAS/SIGNAL/TEST	LRV	URV	ACTION	LRV	URV
	Bench set-Travel					
	Positioner input-Travel					
	Transducer input-Output					
	Test pressure					

COMPONENT IDENTIFICATIONS		
COMPONENT TYPE	MANUFACTURER	MODEL NUMBER

Rev	Date	Revision Description	By	Appv1	Appv2	Appv3	REMARKS

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14		Output signal type		63		Cert/Approval type	
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43		Press port orifice size	Exhaust port	92		Mounting location/type	
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45		Cert/Approval type		94		Diaphragm material	
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48		Body material		97		BOOSTER RELAY	
49		Seat/Disc/Diaphragm matl		98		Configuration type	
50				99		End conn nominal size	Style
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53		Housing type		102		Supply port size	Exh port
54		Sensor style		103		Body material	
55		Output signal type		104		Closure member matl	
56		Accuracy/Linearity rating		105		Diaphragm material	
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111		TAG NO/FUNCTIONAL IDENT	MEAS/SIGNAL/TEST	LRV	URV	ACTION	LRV	URV
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114			Position setpoint-Output					
115			Supply setpoint-Output					
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117								

118		COMPONENT IDENTIFICATIONS		
119		COMPONENT TYPE	MANUFACTURER	MODEL NUMBER
120		Position transmitter		
121		Solenoid/Pilot valve		
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RESPONSIBLE ORGANIZATION	FLOW NOZZLE w/wo METER TUBE Device Specification	SPECIFICATION IDENTIFICATIONS
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BODY		PERFORMANCE CHARACTERISTICS	
Body type		Max press at design temp	At
Fabrication style		Min working temperature	Max
Process conn nominal size	Rating	Uncalibrated accuracy	
End conn termn type	Style	Perm head loss at URL	
Flange facing finish			
Pressure tap quantity			
Press tap conn nom size	Style		
Stampings			
Nozzle material			

THROAT	
Diameter ratio (Beta d/D)	
Throat diameter	

METER TUBE		ACCESSORIES	
Bore type		Purge system	
End conn nominal size	Rating		
End conn termn type	Style		
Flange facing finish			
Upstream length			
Pipe schedule no			
Downstream length			
Coupling quantity			
Coupling nominal size	Rating		
Coupling locations			
Internal bore diameter			
Flow straightener type			
Pipe/Tube material			
Flow straightener matl			
Bolting material			
Exterior coating material			
End termination material			
Gasket material			
Diffuser cone material			

SPECIAL REQUIREMENTS	
Custom tag	
Reference specification	
Special preparation	
Compliance standard	
Construction code	
Calibration report	
Weld radiographs	

PHYSICAL DATA	
Estimated weight	
Face-to-face dimension	
Removal clearance	
Mfr reference dwg	

CALIBRATIONS AND TEST		INPUT OR TEST			OUTPUT	
TAG NO/FUNCTIONAL IDENT	MEAS/SIGNAL/TEST	LRV	URV		LRV	URV
	Flow rate–Diff pressure					
	Test pressure					

COMPONENT IDENTIFICATIONS

COMPONENT TYPE	MANUFACTURER	MODEL NUMBER

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RESPONSIBLE ORGANIZATION	ORIFICE PLATE	SPECIFICATION IDENTIFICATIONS	
	Device Specification	Document no	
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ORIFICE PLATE AND HOLDER	
Plate type	
Holder/Seal unit style	
Nominal size	
API ring no	
Orifice flange nom rating	
Orifice bore type	
Orifice inlet edge style	
Diameter ratio (Beta d/D)	
Orifice bore diameter	
Plate outside diameter	
Plate thickness	
Vent/Drain hole size	
Vent/Drain hole location	
Stampings required	
Plate material	
Seal material	
Holder/Ring material	

ACCESSORIES	
Orifice holding block	
Gasket set	

SPECIAL REQUIREMENTS	
Custom tag	
Reference specification	
Special preparation	
Compliance standard	
Calculation report	

PHYSICAL DATA	
Estimated weight	
Removal clearance	
Mfr reference dwg	

CALIBRATIONS AND TEST		INPUT			OUTPUT	
TAG NO/FUNCTIONAL IDENT	MEAS/SIGNAL/TEST	LRV	URV		LRV	URV
	Flow rate-Diff pressure					

COMPONENT IDENTIFICATIONS		
COMPONENT TYPE	MANUFACTURER	MODEL NUMBER

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RESPONSIBLE ORGANIZATION	ORIFICE PLATE ASSEMBLY w/wo METER TUBE Device Specification	SPECIFICATION IDENTIFICATIONS	
		Document no	
		Latest revision	Date
		Issue status	

ORIFICE FLANGE OR FITTING			
Flange/Fitting type			
Pressure tap style			
End conn nominal size		Rating	
End conn termn type		Style	
Flange facing finish			
End conn schedule no			
Press tap conn nom size		Style	
Pressure tap orientation			
Tap connection plug type			
Centering/Dowel pins			
Jack screw material			
Flange/Fitting material			
Trim material			
Bolting material			
Gasket material			
Lubrication material			

ORIFICE PLATE AND HOLDER			
Plate type			
Holder/Seal unit style			
API ring no		Rating	
Orifice bore type			
Orifice inlet edge style			
Diameter ratio (Beta d/D)			
Orifice bore diameter			
Plate outside diameter			
Plate thickness			
Vent/Drain hole size			
Vent/Drain hole location			
Stampings			
Plate material			
Seal material			
Holder/Ring material			

METER TUBE			
Bore type			
End conn nominal size		Rating	
End conn termn category		Style	
Flange facing finish			
Upstream length			
Pipe schedule no			
Downstream length			
Coupling quantity			
Coupling nominal size		Rating	

METER TUBE Continued	
Coupling locations	
Internal bore diameter	
Flow straightener type	
Pipe/Tube material	
Bolting material	
Exterior coating material	
End termination material	
Gasket material	

PERFORMANCE CHARACTERISTICS			
Max press at design temp		At	
Min working temperature		Max	

ACCESSORIES	
Orifice holding blocks	

SPECIAL REQUIREMENTS	
Custom tag	
Reference specification	
Special preparation	
Compliance standard	
Construction code	

PHYSICAL DATA	
Estimated weight	
Face-to-face dimension	
Overall height	
Removal clearance	
Mfr reference dwg	

CALIBRATIONS AND TEST		INPUT OR TEST			OUTPUT	
TAG NO/FUNCTIONAL IDENT	MEAS/SIGNAL/TEST	LRV	URV		LRV	URV
	Flow rate – Diff pressure					
	Test pressure					

COMPONENT IDENTIFICATIONS

COMPONENT TYPE	MANUFACTURER	MODEL NUMBER

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RESPONSIBLE ORGANIZATION	MULTIVARIABLE FLOW TRANSMITTER	SPECIFICATION IDENTIFICATIONS	
	Device Specification	Document no	
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TRANSMITTER BODY			
Body/Flange type			
Process conn nominal size		Rating	
Process conn termn type		Style	
Vent/Drain location			
Mounting type			
Body/Flange material			
Vent/Drain material			
Bolting material			
Flange adapter material			
Gasket/O ring material			
Mounting kit material			

SENSING ELEMENT			
Detector type			
Min diff pressure span		Max	
Min pressure span		Max	
Temp input signal			
Diaphragm/Wetted material			
Fill fluid material			

TRANSMITTER	
Output signal type	
Enclosure type no/class	
Characteristic curve	
Digital communication std	
Signal power source	
Transient protection	
Integral indicator style	
Signal termination type	
Cert/Approval type	
Span-Zero adjust lct	
Failure/Diagnostic action	
Enclosure material	

PERFORMANCE CHARACTERISTICS			
Max press at design temp		At	
Min working temp		Max	
Diff press accuracy rating			
Press accuracy rating			
Temp accuracy rating			
Diff press LRL		URL	
Press Lower Range-limit		URL	
Fill fluid sp gr		At temp	
Min ambient working temp		Max	

ACCESSORIES	
Remote indicator style	
Heating kit style	
Manifold valve style	
Manifold material	

SPECIAL REQUIREMENTS	
Custom tag	
Reference specification	
Special preparation	
Compliance standard	
Calibration report	
Software configuration	

PHYSICAL DATA			
Estimated weight			
Overall height			
Removal clearance			
Signal conn nominal size		Style	
Mfr reference dwg			

CALIBRATIONS AND TEST		INPUT OR TEST			OUTPUT OR SCALE	
TAG NO/FUNCTIONAL IDENT	MEAS/SIGNAL/SCALE	LRV	URV	ACTION	LRV	URV
	Flow rate-Analog output					
	Flow rate-Scale					
	Diff Press-Digital output					
	Press-Digital output					
	Temp-Digital output					
	Flow rate-Digital output					

COMPONENT IDENTIFICATIONS		
COMPONENT TYPE	MANUFACTURER	MODEL NUMBER

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RESPONSIBLE ORGANIZATION	VENTURI OR FLOW TUBE w/wo METER TUBE Device Specification	SPECIFICATION IDENTIFICATIONS
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BODY OR TUBE			
Tube form			
Fabrication style			
Inlet conn nominal size		Rating	
Outlet conn nominal size		Rating	
End conn termn type		Style	
Flange facing finish			
Pressure tap quantity			
Press tap conn nom size		Rating	
Press tap conn type		Style	
Press tap orientation			
Body/Tube material			
Bolting material			
Exterior coating material			
End termination material			
Gasket material			

THROAT	
Mounting type	
Diameter ratio (beta d/D)	
Throat diameter	
Cone material	
Throat material	

METER TUBE			
Bore type			
End conn nominal size		Rating	
End conn termn type		Style	
Flange facing finish			
Upstream length			
Pipe schedule no			
Downstream length			
Coupling quantity			
Coupling nominal size		Rating	
Coupling locations			
Internal bore diameter			
Flow straightener type			
Pipe/Tube material			
Bolting material			
Exterior coating material			
End termination material			
Gasket material			

PERFORMANCE CHARACTERISTICS			
Max press at design temp		At	
Min working temperature		Max	
Uncalibrated accuracy			
Perm head loss at URL			

ACCESSORIES	
Purge system	
Cleaner style	
Valve style	
Valves and Nipple matl	

SPECIAL REQUIREMENTS	
Custom tag	
Reference specification	
Special preparation	
Compliance standard	
Construction code	
Calibration report	
Weld radiographs	

PHYSICAL DATA	
Estimated weight	
Face-to-face dimension	
Removal clearance	
Mfr reference dwg	

CALIBRATIONS AND TEST		INPUT OR TEST			OUTPUT	
TAG NO/FUNCTIONAL IDENT	MEAS/SIGNAL/TEST	LRV	URV		LRV	URV
	Flow rate – Diff pressure					
	Test pressure					

COMPONENT IDENTIFICATIONS		
COMPONENT TYPE	MANUFACTURER	MODEL NUMBER

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RESPONSIBLE ORGANIZATION	MAGNETIC FLOWMETER w/wo INTEGRAL TOTALIZER INDICATOR Device Specification	SPECIFICATION IDENTIFICATIONS	
		Document no	
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FLOWMETER BODY			
Body type			
Flow tube style			
End conn nominal size		Rating	
End conn termn type		Style	
Flow tube diameter		Thickness	
Hardware mounting kit			
Flow tube material			
Lining material			
End termination material			
Gnd/protective ring matl			

END EXTENSIONS			
End termination type		Style	
Bolting material			
End termination material			
Gasket/O ring material			

SENSING ELEMENT	
Electrode type	
Insertion length	
Electrode material	

COILS AND HOUSING	
Housing construction type	
Coil conn arrangement	
Enclosure type no/class	
Signal power source	
Signal termination type	
Cert/Approval type	
Housing material	

TRANSMITTER OR CONVERTER	
Housing type	
Output signal type	
Enclosure type no/class	
Characteristic curve	
Digital communication std	
Signal power source	
Failsafe style	
Integral indicator style	
Signal termination type	
Cert/Approval type	
Mounting location/type	
Failure/Diagnostic action	
Enclosure material	

TOTALIZER INDICATOR			
Totalizer type			
Enclosure type no/class			
Signal power source			
Contacts arrangement		Quantity	
Totalizer reset style			
Integral indicator style			
Cert/Approval type			
Mounting location/type			
Enclosure material			

PERFORMANCE CHARACTERISTICS			
Min press at design temp		At	
Max press at design temp			
Min working temperature		Max	
Accuracy rating			
Min velocity URL		Max	
Min liquid conductivity			
Output signal damping LRL		URL	
Min ambient working temp		Max	
Contacts ac rating		At max	
Contacts dc rating		At max	
Max sensor to receiver lg			

ACCESSORIES	
Connecting cables length	
Cable Glands	
Ultrasonic cleaner style	
Empty tube detector	
Calibrator adaptor	
Calibrator/configurator	

SPECIAL REQUIREMENTS	
Custom tag	
Reference specification	
Compliance standard	
Calibration report	
Software configuration	

PHYSICAL DATA			
Estimated weight			
Face-to-face dimension			
Overall height			
Removal clearance			
Signal conn nominal size		Style	
Mfr reference dwg			

CALIBRATIONS AND TEST		INPUT OR TEST			OUTPUT OR SCALE	
TAG NO/FUNCTIONAL IDENT	MEAS/SIGNAL/TEST	LRV	URV	ACTION	LRV	URV
	Flow rate-Analog output					
	Flow rate-Digital output					
	Flow rate-Freq output					
	Flow rate-Scale					
	Test pressure					

COMPONENT IDENTIFICATIONS		
COMPONENT TYPE	MANUFACTURER	MODEL NUMBER

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RESPONSIBLE ORGANIZATION	VORTEX OR SWIRL FLOWMETER w/wo TOTALIZER INDICATOR Device Specification	SPECIFICATION IDENTIFICATIONS
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FLOWMETER BODY			
Body type			
End conn nominal size		Rating	
End conn termn type		Style	
Seal type			
Mounting hardware			
Body wetted material			
End termination material			
Gasket/Seal material			

SENSING ELEMENT	
Sensor type	
Sensor wetted material	
Fill fluid material	

CONNECTION HEAD	
Housing type	
Enclosure type no/class	
Signal termination type	
Cert/Approval type	
Enclosure material	

TRANSMITTER OR COMPUTER	
Housing type	
Measurement compensation	
Output signal type	
Enclosure type no/class	
Span-zero adjustment	
Characteristic curve	
Digital communication std	
Signal power source	
Failsafe style	
Integral indicator style	
Signal termination type	
Cert/Approval type	
Mounting location/type	
Calibration type	
Failure/Diagnostic action	
Enclosure material	

TOTALIZER INDICATOR			
Totalizer type			
Enclosure type no/class			
Signal power source			
Contacts arrangement		Quantity	
Totalizer reset style			
Integral indicator style			
Cert/Approval type			
Mounting location/type			
Enclosure material			

PERFORMANCE CHARACTERISTICS			
Max press at design temp		At	
Min working temperature		Max	
Flow rate accuracy rating			
Density accuracy rating			
Min velocity URL		Max	
Density LRL		URL	
Min reqd back press			
Pressure drop at flow URL			
Min ambient working temp		Max	
Contacts ac rating		At max	
Contacts dc rating		At max	
Max sensor to receiver lg			

ACCESSORIES	
Connecting cables length	
Isolation manifold	
Remote indicator style	
Calibrator adapter	

SPECIAL REQUIREMENTS	
Custom tag	
Reference specification	
Special preparation	
Compliance standard	
Calibration report	
Software configuration	

PHYSICAL DATA			
Estimated weight			
Face-to-face dimension			
Overall height			
Removal clearance			
Signal conn nominal size		Style	
Mfr reference dwg			

CALIBRATIONS AND TEST		INPUT OR TEST			OUTPUT OR SCALE	
TAG NO/FUNCTIONAL IDENT	MEAS/SIGNAL/TEST	LRV	URV	ACTION	LRV	URV
	Flow rate-Analog output					
	Mass flow-Analog output					
	Meas-Analog output					
	Meas-Freq output 1					
	Meas-Freq output 2					
	Measurement-Scale					
	Temp-Digital output					
	Meas-Digital output					
	Density-Digital output					
	Test pressure					

COMPONENT IDENTIFICATIONS		
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RESPONSIBLE ORGANIZATION	CORIOLIS MASS FLOWMETER w/wo TOTALIZER INDICATOR Device Specification	SPECIFICATION IDENTIFICATIONS	
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FLOWMETER BODY AND HOUSING			
Body/Housing type			
End conn nominal size		Rating	
End conn termn type		Style	
Body wetted material			
Housing material			
End termination material			

FLOWTUBE ASSEMBLY	
Flowtube type	
Flowtube diameter	
Flowtube material	

CONNECTION HEAD	
Housing type	
Enclosure type no/class	
Signal termination type	
Cert/Approval type	
Housing material	

TRANSMITTER OR COMPUTER	
Housing type	
Measurement compensation	
Output signal type	
Enclosure type no/class	
Span-zero adjustment	
Characteristic curve	
Digital communication std	
Signal power source	
Failsafe style	
Integral indicator style	
Signal termination type	
Cert/Approval type	
Mounting location/type	
Failure/Diagnostic action	
Enclosure material	

TOTALIZER INDICATOR			
Totalizer type			
Enclosure type no/class			
Signal power source			
Contacts arrangement		Quantity	
Totalizer reset style			
Integral indicator style			
Cert/Approval type			
Mounting location/type			
Enclosure material			

PERFORMANCE CHARACTERISTICS			
Max press at design temp		At	
Min working temperature		Max	
Flow rate accuracy rating			
Density accuracy rating			
Min flow URL		Max	
Density LRL		URL	
Sec enclosure press rating			
Pressure drop at flow URL			
Min ambient working temp		Max	
Contacts ac rating		At max	
Contacts dc rating		At max	
Max sensor to receiver lg			

ACCESSORIES	
Connecting cables length	
Heating kit style	
Calibrator/configurator	

SPECIAL REQUIREMENTS	
Custom tag	
Reference specification	
Special preparation	
Compliance standard	
Calibration report	
Software configuration	

PHYSICAL DATA			
Estimated weight			
Face-to-face dimension			
Overall height			
Removal clearance			
Signal conn nominal size		Style	
Mfr reference dwg			

CALIBRATIONS AND TEST		INPUT OR TEST			OUTPUT OR SCALE	
TAG NO/FUNCTIONAL IDENT	MEAS/SIGNAL/TEST	LRV	URV	ACTION	LRV	URV
	Mass flow-Analog output					
	Meas-Analog output 2					
	Meas-Analog output 3					
	Meas-Freq output 1					
	Meas-Freq output 2					
	Measurement–Scale					
	Temp-Digital output					
	Meas-Digital output					
	Density-Digital output					
	Test pressure					

COMPONENT IDENTIFICATIONS		
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RESPONSIBLE ORGANIZATION	CAPACITANCE OR RF ADMITTANCE LEVEL TRANSMITTER w/wo SWITCHES Device Specification	SPECIFICATION IDENTIFICATIONS
		Document no Latest revision / Date Issue status

PROCESS CONNECTION			
Body/Fitting type			
Process conn nominal size		Rating	
Process conn termn type		Style	
Wetted material			
Flange material			
Seal/O ring material			

SENSING ELEMENT	
Configuration type	
Probe diameter	
Insertion length	
Inactive length	
Integral cable length	
Probe material	
Insulation/Sheath matl	
Weight material	

CONNECTION HEAD OR PREAMPLIFIER	
Type	
Measurement compensation	
Input signal URL	
Output signal type	
Enclosure type no/class	
Power source	
Cert/Approval type	
Mounting location/type	
Enclosure material	

TRANSMITTER w/wo SWITCHES			
Housing type			
Measurement compensation			
Output signal type			
Input signal URL			
Enclosure type no/class			
Characteristic curve			
Signal power source			
Contact arrangement		Quantity	
Failsafe style			
RFI protection			
Integral indicator style			
Cert/Approval type			
Mounting location/type			
Static discharge prot			
Failure/Diagnostic action			
Adjustable signal damping			

TRANSMITTER w/wo SWITCHES continued			
Switch time delay			
Enclosure material			

PERFORMANCE CHARACTERISTICS			
Max press at design temp		At	
Min working temperature		Max	
Max discharge to sensor			
Accuracy rating			
Min measurement span		Max	
Min ambient working temp		Max	
Contacts ac rating		At max	
Contacts dc rating		At max	
Max sensor to receiver lg			

ACCESSORIES	
Connecting cables length	
Power supply	
Local indicator style	
Intrinsic safety barrier	

SPECIAL REQUIREMENTS	
Custom tag	
Reference specification	
Compliance standard	
Calibration report	

PHYSICAL DATA			
Estimated weight			
Overall height			
Removal clearance			
Signal conn nominal size		Style	
Mfr reference dwg			

CALIBRATIONS AND TEST		INPUT OR SETPOINT			OUTPUT OR SCALE	
TAG NO/FUNCTIONAL IDENT	MEAS/SIGNAL/TEST	LRV	URV	ACTION	LRV	URV
	Level-Analog output					
	Level-Scale					
	Level setpoint 1-Output					
	Level setpoint 2-Output					
	Level setpoint 3-Output					

COMPONENT IDENTIFICATIONS		
COMPONENT TYPE	MANUFACTURER	MODEL NUMBER

Rev	Date	Revision Description	By	Appv1	Appv2	Appv3	REMARKS

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Form: 20L2101 Rev 0

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RESPONSIBLE ORGANIZATION	DIFFERENTIAL PRESSURE LEVEL TRANSMITTER - FLANGE MOUNTED Device Specification	SPECIFICATION IDENTIFICATIONS	
		Document no	
		Latest revision	Date
		Issue status	

TRANSMITTER BODY			
Body/Flange type			
Process conn nominal size	Rating		
Process conn termn type	Style		
Vent/Drain location			
Mounting type			
Body/Flange material			
Vent/Drain material			
Bolting material			
Flange adapter material			
Gasket/O ring material			
Mounting kit material			

SENSING ELEMENT			
Detector type			
Min diff pressure span	Max		
Diaphragm/Wetted material			
Fill fluid material			

TRANSMITTER			
Output signal type			
Enclosure type no/class			
Characteristic curve			
Digital communication std			
Signal power source			
Transient protection			
Integral indicator style			
Signal termination type			
Cert/Approval type			
Span-Zero adjust loc			
Failure/Diagnostic action			
Enclosure material			

INTEGRAL SEAL			
Seal type			
Process conn nominal size	Rating		
Process conn termn type	Style		
Diaphragm extension lg			
Diaphragm material			
Upper housing/Flange matl			

REMOTE SEAL			
Seal type			
Process conn nominal size	Rating		
Process conn termn type	Style		
Diaphragm extension lg			
Instrument conn nom size			
Flushing conn quantity			
Capillary/Fitting dia	Length		
Diaphragm material			
Capillary-armor material			
Bolting material			
Upper housing material			
Lower housing/Flange matl			
Gasket/Oring material			
Fill fluid material			

PERFORMANCE CHARACTERISTICS			
Max press at design temp	At		
Min working temp	Max		
Accuracy rating			
Diff pressure LRL	URL		
Fill fluid sp gr	At temp		
Ambient temperature error			
Min ambient working temp	Max		

ACCESSORIES			
Air set filter style			
Air set gauges			
Heating kit style			
Remote indicator style			

SPECIAL REQUIREMENTS			
Custom tag			
Reference specification			
Special preparation			
Compliance standard			
Software configuration			

PHYSICAL DATA			
Estimated weight			
Overall height			
Removal clearance			
Signal conn nominal size	Style		
Mfr reference dwg			

CALIBRATIONS AND TEST		INPUT OR TEST			OUTPUT OR SCALE	
TAG NO/FUNCTIONAL IDENT	MEAS/SIGNAL/TEST	LRV	URV	ACTION	LRV	URV
	Diff press-Analog output					
	Diff press-Scale					
	Diff press-Digital output					
	Press-Digital output					
	Temp-Digital output					

COMPONENT IDENTIFICATIONS		
COMPONENT TYPE	MANUFACTURER	MODEL NUMBER

Rev	Date	Revision Description	By	Appv1	Appv2	Appv3	REMARKS

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1	RESPONSIBLE ORGANIZATION		DISPLACER-TYPE LEVEL TRANSMITTER OR LOCAL CONTROLLER Device Specification		6	SPECIFICATION IDENTIFICATIONS	
2					7	Document no	
3					8	Latest revision	Date
4					9	Issue status	
5					10		

11	BODY OR CAGE			60	TRANSMITTER OR CONTROLLER continued		
12	Body/Cage type			61	Cert/Approval type		
13	Process conn nominal size	Rating		62	Mounting location/type		
14	Process conn termn type	Style		63	Span-Zero adjust lct		
15	Flange facing finish			64	Enclosure material		
16	Lower conn location	Upper		65			
17	Gage glass conn nom size	Rating		66			
18	Gage glass conn type	Style		67			
19	Vent/Drain conn nom size	Rating		68	PERFORMANCE CHARACTERISTICS		
20	Vent/Drain termn type	Style		69	Max press at design temp	At	
21	Head orientation/type			70	Min working temperature	Max	
22	Extension/Heat insulator			71	Accuracy rating		
23	Body/Cage material			72	Level Lower Range-Limit	URL	
24	Bolting material			73	Min level span	Max	
25	Gasket/O ring material			74	Min differential sp gr		
26				75	Min ambient working temp	Max	
27				76	Contacts ac rating	At max	
28				77	Contacts dc rating	At max	
29	SENSING ELEMENT			78			
30	Sensor type			79			
31	Sp gr Lower Range-Limit	URL		80			
32	Displacer diameter	Length		81	ACCESSORIES		
33	Extension length			82	Connecting cables length		
34	Displacer material			83	Air set filter style		
35	Torque tube/Spring matl			84	Air set gauges		
36	Extension/Cable material			85	Gage glass style		
37	Trim material			86	Gage valve style		
38				87			
39				88			
40	CONNECTION HEAD			89	SPECIAL REQUIREMENTS		
41	Type			90	Custom tag		
42	Enclosure type no/class			91	Reference specification		
43	Signal termination type			92	Special preparation		
44	Cert/Approval type			93	Compliance standard		
45	Enclosure material			94	Construction code		
46				95	Software configuration		
47				96			
48				97			
49	TRANSMITTER OR CONTROLLER w/wo SWITCHES			98	PHYSICAL DATA		
50	Housing type			99	Estimated weight		
51	Output signal type			100	Overall height		
52	Enclosure type no/class			101	Removal clearance		
53	Control mode			102	Upper to lower conn lg		
54	Digital communication std			103	Lower to drain conn lg		
55	Signal power source			104	Signal conn nominal size	Style	
56	Contacts arrangement	Quantity		105	Mfr reference dwg		
57	Transient protection			106			
58	Integral indicator style			107			
59	Signal termination type			108			

110	CALIBRATIONS AND TEST		INPUT OR SETPOINT			OUTPUT OR SCALE	
111	TAG NO/FUNCTIONAL IDENT	MEAS/SIGNAL/TEST	LRV	URV	ACTION	LRV	URV
112		Level-Analog output					
113		Level-Digital output					
114		Level-Scale					
115		Level setpoint 1-Output					
116		Level setpoint 2-Output					
117		Test pressure					

118	COMPONENT IDENTIFICATIONS		
119	COMPONENT TYPE	MANUFACTURER	MODEL NUMBER
120			
121			
122			
123			
124			
125			

Rev	Date	Revision Description	By	Appv1	Appv2	Appv3	REMARKS

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RESPONSIBLE ORGANIZATION	NUCLEAR RADIATION LEVEL TRANSMITTER w/wo SWITCHES Device Specification	SPECIFICATION IDENTIFICATIONS
		Document no
		Latest revision / Date
		Issue status

RADIATION SOURCE	
Source type	
Number of sources	
Source size	
Fireproof design type	
Shutter style	
Position switch type	
Mounting location/type	
Collimation angle	
Source material	
Housing material	

SENSING ELEMENT	
Detector type	
Detector style	
Number of detectors	
Active length	

CONNECTION HEAD OR PREAMPLIFIER	
Type	
Housing type	
Measurement compensation	
Output signal type	
Enclosure type no/class	
Signal power source	
Cert/Approval type	
Mounting location/type	
Enclosure material	

TRANSMITTER w/wo SWITCHES	
Housing type	
Measurement compensation	
Output signal type	
Enclosure type no/class	
Characteristic curve	
Digital communication std	
Signal power source	
Contact arrangement	Quantity
Failsafe style	
Integral indicator style	
Cert/Approval type	
Mounting location/type	
Failure/Diagnostic action	
Enclosure material	

PERFORMANCE CHARACTERISTICS	
Level accuracy rating	
Density accuracy rating	
Min measurement span	Max
Response time	
Min ambient working temp	Max
Contacts ac rating	At Max
Contacts dc rating	At Max
Max sensor to receiver lg	

ACCESSORIES	
Connecting cables length	
Mounting hardware	
Calibrator	

SPECIAL REQUIREMENTS	
Custom tag	
Reference specification	
Compliance standard	

PHYSICAL DATA	
Estimated weight	
Overall length	
Removal clearance	
Signal conn nominal size	Style
Mfr reference dwg	

CALIBRATIONS AND TEST		INPUT OR SETPOINT			OUTPUT OR SCALE	
TAG NO/FUNCTIONAL IDENT	MEAS/SIGNAL/TEST	LRV	URV	ACTION	LRV	URV
	Level-Analog output					
	Level-Digital output					
	Level-Scale					
	Density-Digital output					
	Level setpoint 1-Output					
	Level setpoint 2-Output					

COMPONENT IDENTIFICATIONS		
COMPONENT TYPE	MANUFACTURER	MODEL NUMBER

Rev	Date	Revision Description	By	Appv1	Appv2	Appv3	REMARKS

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1		RESPONSIBLE ORGANIZATION	RESISTANCE-TAPE LEVEL TRANSMITTER w/wo SWITCHES Device Specification		6		SPECIFICATION IDENTIFICATIONS
2					7		Document no
3					8		Latest revision / Date
4					9		Issue status
5					10		

11		PROCESS CONNECTION		60		PERFORMANCE CHARACTERISTICS	
12		Body/Fitting type		61		Max press at design temp	At
13		Process conn nominal size	Rating	62		Min working temperature	Max
14		Process conn termn type	Style	63		Repeatability	
15		Shell/cap material		64		Min measurement span	Max
16		Seal/O ring material		65		Sensor resistance coef	
17				66		Actuation depth	
18				67		Min ambient working temp	Max
19				68		Contacts ac rating	At max
20		SENSING ELEMENT		69		Contacts dc rating	At max
21		Detector type		70		Max sensor to receiver lg	
22		Temperature sensor qty		71			
23		Temperature coefficient		72			
24		Temp nominal resistance	At	73			
25		Sheath style		74			
26		Overall length		75			
27		Inactive length		76			
28		Sheath material		77			
29		Protector material		78			
30				79			
31				80			
32				81			
33		CONNECTION HEAD OR HOUSING		82			
34		Type		83			
35		Housing style		84			
36		Enclosure type no/class		85		ACCESSORIES	
37		Signal termination type		86		Connecting cables length	
38		Cert/Approval type		87		Local indicator style	
39		Mounting location/type		88		Intrinsic safety barrier	
40		Enclosure material		89			
41		Breather material		90			
42		Fitting material		91			
43				92			
44				93		SPECIAL REQUIREMENTS	
45		TRANSMITTER w/wo SWITCHES		94		Custom tag	
46		Housing type		95		Reference specification	
47		Output signal type		96		Compliance standard	
48		Enclosure type no/class		97			
49		Characteristic curve		98			
50		Signal power source		99			
51		Contacts arrangement	Quantity	100		PHYSICAL DATA	
52		Failsafe style		101		Estimated weight	
53		Integral indicator style		102		Overall height	
54		Cert/Approval type		103		Removal clearance	
55		Mounting location/type		104		Signal conn nominal size	Style
56		Enclosure material		105		Mfr reference dwg	
57				106			
58				107			
59				108			

110		CALIBRATIONS AND TEST		INPUT OR SETPOINT			OUTPUT OR SCALE	
111		TAG NO/FUNCTIONAL IDENT	MEAS/SIGNAL/TEST	LRV	URV	ACTION	LRV	URV
112			Level-Analog output					
113			Temp-Analog output					
114			Level-Scale					
115			Level setpoint 1-Output					
116			Level setpoint 2-Output					
117								

118		COMPONENT IDENTIFICATIONS		
119		COMPONENT TYPE	MANUFACTURER	MODEL NUMBER
120				
121				
122				
123				
124				
125				

Rev	Date	Revision Description	By	Appv1	Appv2	Appv3	REMARKS

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		Document no
		Latest revision / Date
		Issue status

PROCESS CONNECTION			
Body/Fitting type			
Process conn nominal size		Rating	
Process conn termn type		Style	
Wetted material			
Flange material			
Seal/O ring material			

SENSING ELEMENT	
Detector type	
Detector style	
Insertion length	
Integral cable length	

CONNECTION HEAD	
Type	
Enclosure type no/class	
Cert/Approval type	
Mounting location/type	
Enclosure material	

TRANSMITTER w/wo SWITCHES			
Housing type			
Measurement compensation			
Output signal type			
Enclosure type no/class			
Control sequence			
Characteristic curve			
Digital communication std			
Signal power source			
Contact arrangement		Quantity	
Failsafe style			
Integral indicator style			
Cert/Approval type			
Mounting location/type			
Failure/Diagnostic action			
Signal processing			
Enclosure material			

PERFORMANCE CHARACTERISTICS			
Max press at design temp		At	
Min working temperature		Max	
Accuracy rating			
Level lower range limit		URL	
Min measurement span		Max	
Beam angle			
Min ambient working temp		Max	
Contacts ac rating		At max	
Contacts dc rating		At max	
Max sensor to receiver lg			

ACCESSORIES	
Connecting cables length	
Enclosure heater	
Mounting hardware	

SPECIAL REQUIREMENTS	
Custom tag	
Reference specification	
Compliance standard	
Calibration report	

PHYSICAL DATA			
Estimated weight			
Overall height			
Removal clearance			
Signal conn nominal size		Style	
Mfr reference dwg			

CALIBRATIONS AND TEST		INPUT OR SETPOINT			OUTPUT OR SCALE	
TAG NO/FUNCTIONAL IDENT	MEAS/SIGNAL/TEST	LRV	URV	ACTION	LRV	URV
	Level–Analog output					
	Level-Scale					
	Level setpoint 1-Output					
	Level setpoint 2-Output					
	Level setpoint 3-Output					

COMPONENT IDENTIFICATIONS		
COMPONENT TYPE	MANUFACTURER	MODEL NUMBER

Rev	Date	Revision Description	By	Appv1	Appv2	Appv3	REMARKS

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RESPONSIBLE ORGANIZATION	CAPACITANCE OR RF ADMITTANCE LEVEL SWITCH Device Specification	SPECIFICATION IDENTIFICATIONS
		Document no
		Latest revision / Date
		Issue status

PROCESS CONNECTION	
Body/Fitting type	
Process conn nominal size	Rating
Process conn termn type	Style
Wetted material	
Flange material	
Seal/O ring material	

SENSING ELEMENT	
Configuration type	
Number of elements	
Probe diameter	
Insertion length	
Inactive/Dead length	
Integral cable length	
Probe material	
Insulation/Sheath matl	
Weight material	

CONNECTION HEAD OR PREAMPLIFIER	
Type	
Measurement compensation	
Input signal URL	
Output signal type	
Enclosure type no/class	
Signal power source	
Cert/Approval type	
Mounting location/type	
Enclosure material	

SWITCH MECHANISM	
Housing type	
Element style	
Measurement compensation	
Output signal type	
Input signal LRL	URL
Enclosure type no/class	
Signal power source	
Dead band type	
Contacts arrangement	Quantity
Failsafe style	
Integral indicator style	
Cert/Approval type	
Mounting location/type	
Static discharge prot	
Failure/Diagnostic action	

SWITCH MECHANISM continued	
RFI protection	
Switch time delay	
Enclosure material	

PERFORMANCE CHARACTERISTICS	
Max press at design temp	At
Min working temperature	Max
Max discharge to sensor	
Repeatability/Sensitivity	
Min measurement span	Max
Min ambient working temp	Max
Contacts ac rating	At max
Contacts dc rating	At max
Max sensor to receiver lg	

ACCESSORIES	
Connecting cables length	
Mounting hardware	
External relay	
Intrinsic safety barrier	

SPECIAL REQUIREMENTS	
Custom tag	
Reference specification	
Compliance standard	

PHYSICAL DATA	
Estimated weight	
Overall height	
Removal clearance	
Signal conn nominal size	Style
Mfr reference dwg	

CALIBRATIONS AND TEST		SETPOINT			OUTPUT	
TAG NO/FUNCTIONAL IDENT	MEAS/SIGNAL/TEST	LRV	URV	ACTION	LRV	URV
	Level setpoint 1-Output					
	Level setpoint 2-Output					
	Level setpoint 3-Output					
	Level setpoint 4-Output					

COMPONENT IDENTIFICATIONS

COMPONENT TYPE	MANUFACTURER	MODEL NUMBER

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RESPONSIBLE ORGANIZATION	FLOAT OR DISPLACER LEVEL SWITCH	SPECIFICATION IDENTIFICATIONS	
	Device Specification	Document no	
		Latest revision	Date
		Issue status	

BODY OR CAGE			
Body/Cage type			
Process conn nominal size		Rating	
Process conn termn type		Style	
Flange facing finish			
Lower conn location		Upper	
Gage glass conn nom size		Style	
Try-cock conn nom size		Style	
Head orientation/type			
Extension/Heat insulator			
Body/Cage material			
Bolting material			
Gasket material			

SENSING ELEMENT			
Sensor type			
Sp gr Lower Range-Limit		URL	
Adjustable LRL		URL	
Sensor quantity			
Float/Displacer dia		Length	
Extension/stem length			
Float/Displacer material			
Torque tube/Spring matl			
Magnetic sleeve material			
Extension/Cable material			
Trim material			

SWITCH MECHANISM			
Housing type			
Element style			
Output signal type			
Enclosure type no/class			
Control sequence			
Signal power source			
Contacts arrangement		Quantity	
Signal termination type			
Cert/Approval type			
Dead band type			
Enclosure material			
Exterior coating material			

PERFORMANCE CHARACTERISTICS			
Max press at design temp		At	
Min working temperature		Max	
Repeatability			
Level Lower Range-Limit		URL	
Dead band rating			
Min ambient working temp		Max	
Contacts ac rating		At max	
Contacts dc rating		At max	

ACCESSORIES	
Try-cocks style	
Gage glass style	
Gage valve style	
Mechanical test mechanism	

SPECIAL REQUIREMENTS	
Custom tag	
Reference specification	
Special preparation	
Compliance standard	
Construction code	

PHYSICAL DATA			
Estimated weight			
Overall height			
Removal clearance			
Upper to lower conn lg			
Lower to drain conn lg			
Signal conn nominal size		Style	
Mfr reference dwg			

CALIBRATIONS AND TEST		SETPOINT			OUTPUT	
TAG NO/FUNCTIONAL IDENT	MEAS/SIGNAL/TEST	LRV	URV	ACTION	LRV	URV
	Level setpoint 1-Output					
	Level setpoint 2-Output					
	Level setpoint 3-Output					
	Level setpoint 4-Output					

COMPONENT IDENTIFICATIONS

COMPONENT TYPE	MANUFACTURER	MODEL NUMBER

Rev	Date	Revision Description	By	Appv1	Appv2	Appv3	REMARKS

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RESPONSIBLE ORGANIZATION	TANK LEVEL GAUGE OR INDICATOR w/wo SWITCHES Device Specification	SPECIFICATION IDENTIFICATIONS	
		Document no	
		Latest revision	Date
		Issue status	

PROCESS CONNECTION			
Body/Fitting type			
Process conn nominal size		Rating	
Process conn termn type		Style	
Wetted material			

SENSING ELEMENT	
Sensor type	
Nominal size	
Sensor material	
Guide cable material	
Bottom anchor material	
Top anchor material	

TAPE CONDUIT	
Sheave elbow type	
Liquid seal style	
Tape material	
Sheave elbow material	
Pipe material	
Liquid seal sheave matl	
Tape wipe material	
Conduit vent fitting matl	

GAUGE OR INDICATOR	
Type	
Spring Motor style	
Level URL	
Seal type	
Graduations and color	
Mounting location/type	
Sensor check style	
Housing material	

SWITCH MECHANISM			
Housing type			
Output signal type			
Max level span			
Enclosure type no/class			
Signal power source			
Contact arrangement		Quantity	
Failsafe style			
Integral indicator style			
Cert/Approval type			

SWITCH MECHANISM continued	
Mounting location/type	
Enclosure material	

PERFORMANCE CHARACTERISTICS			
Max press at design temp		At	
Min working temperature		Max	
Min specific gravity		Max	
Indication accuracy			
Switch repeatability			
Min differential sp gr			
Min ambient working temp		Max	
Contacts ac rating		At max	
Contacts dc rating		At max	

ACCESSORIES	
Calibrator	
Condensate reservoir	
Shock absorber	
Isolation valve	

SPECIAL REQUIREMENTS	
Custom tag	
Reference specification	
Compliance standard	

PHYSICAL DATA			
Estimated weight			
Overall height			
Removal clearance			
Signal conn nominal size		Style	
Mfr reference dwg			

CALIBRATIONS AND TEST		INPUT OR SETPOINT			OUTPUT OR SCALE	
TAG NO/FUNCTIONAL IDENT	MEAS/SIGNAL/TEST	LRV	URV	ACTION	LRV	URV
	Level-Scale					
	Level setpoint 1-Output					
	Level setpoint 2-Output					
	Level setpoint 3-Output					
	Level setpoint 4-Output					

COMPONENT IDENTIFICATIONS		
COMPONENT TYPE	MANUFACTURER	MODEL NUMBER

Rev	Date	Revision Description	By	Appv1	Appv2	Appv3	REMARKS

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		Document no	
	Device Specification	Latest revision	Date
		Issue status	

RADIATION SOURCE	
Shutter type	
Number of sources	
Source size	
Fireproof design type	
Position switch type	
Mounting location/type	
Source material	
Housing material	

SENSING ELEMENT	
Detector type	
Number of detectors	
Active length	
Check source actuation	

CONNECTION HEAD OR PREAMPLIFIER	
Type	
Housing type	
Output signal type	
Enclosure type no/class	
Signal power source	
Cert/approval type	
Mounting location/type	
Enclosure material	

SWITCH MECHANISM			
Housing type			
Output signal type			
Enclosure type no/class			
Signal power source			
Contacts arrangement		Quantity	
Failsafe style			
Integral indicator style			
Cert/Approval type			
Mounting location/type			
Failure/Diagnostics action			
Enclosure material			

PERFORMANCE CHARACTERISTICS			
Repeatability/Sensitivity			
Response time			
Min ambient working temp		Max	
Contacts ac rating		At max	
Contacts dc rating		At max	
Max sensor to receiver lg			

ACCESSORIES	
Connecting cables length	
Mounting hardware	

SPECIAL REQUIREMENTS	
Custom tag	
Reference specification	
Compliance standard	

PHYSICAL DATA			
Estimated weight			
Overall length			
Removal clearance			
Signal conn nominal size		Style	
Mfr reference dwg			

CALIBRATIONS AND TEST		SETPOINT			OUTPUT	
TAG NO/FUNCTIONAL IDENT	MEAS/SIGNAL/TEST	LRV	URV	ACTION	LRV	URV
	Level setpoint 1-Output					
	Level setpoint 2-Output					
	Level setpoint 3-Output					
	Level setpoint 4-Output					

COMPONENT IDENTIFICATIONS		
COMPONENT TYPE	MANUFACTURER	MODEL NUMBER

Rev	Date	Revision Description	By	Appv1	Appv2	Appv3	REMARKS

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RESPONSIBLE ORGANIZATION	ULTRASONIC CONTACT-TYPE LEVEL SWITCH Device Specification	SPECIFICATION IDENTIFICATIONS
		Document no
		Latest revision / Date
		Issue status

PROCESS CONNECTION		SENSING ELEMENT		CONNECTION HEAD OR PREAMPLIFIER	
Body/Fitting type		Configuration type		Type	
Process conn nominal size	Rating	Nominal temp rating		Output signal type	
Process conn termn type	Style	Number of sensors		Enclosure type no/class	
Wetted material		Probe diameter		Signal power source	
Flange/Fitting material		Insertion length		Gain type	
Seal/O ring material		Integral cable length		Cert/Approval type	
		Self test type		Mounting location/type	
		Cert/Approval type		Failure/Diagnostic action	
				Enclosure material	

SWITCH MECHANISM		SWITCH MECHANISM continued		PERFORMANCE CHARACTERISTICS	
Housing type		Enclosure material		Max press at design temp	At
Element style		Exterior coating material		Min working temperature	Max
Input signal type				Repeatability/Sensitivity	
Output signal type				Min ambient working temp	Max
Enclosure type no/class				Contacts ac rating	At Max
Control logic				Contacts dc rating	At Max
Gain type				Max sensor to receiver lg	
Signal power source					
Contacts arrangement	Quantity				
Failsafe style					
Integral indicator style					
Cert/Approval type					
Mounting location/type					
Failure/Diagnostic action					
Switch time delay					

ACCESSORIES		SPECIAL REQUIREMENTS		PHYSICAL DATA	
Connecting cables length		Custom tag		Estimated weight	
Mounting hardware		Reference specification		Overall height	
External relay		Compliance standard		Removal clearance	
Intrinsic safety barrier				Signal conn nominal size	Style
				Mfr reference dwg	

CALIBRATIONS AND TEST		SETPOINT			OUTPUT	
TAG NO/FUNCTIONAL IDENT	MEAS/SIGNAL/TEST	LRV	URV	ACTION	LRV	URV
	Lg conn to sp 1-Output					
	Lg conn to sp 2-Output					
	Lg conn to sp 3-Output					
	Lg conn to sp 4-Output					

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RESPONSIBLE ORGANIZATION	ROTARY LEVEL SWITCH	SPECIFICATION IDENTIFICATIONS
	Device Specification	Document no
		Latest revision / Date
		Issue status

PROCESS CONNECTION			
Body type			
Process conn nominal size		Rating	
Process conn termn type		Style	
Wetted material			
Seal/O ring material			

SENSING ELEMENT	
Sensor type	
Nominal size	
Shaft coupling type	
Shaft guard style	
Extension shaft length	
Sensor material	
Extension/Cable material	
Shaft material	

SWITCH MECHANISM			
Housing type			
Motor style			
Output signal type			
Enclosure type no/class			
Signal power source			
Contacts arrangement		Quantity	
Failsafe style			
Integral indicator style			
Cert/Approval type			
Failure/Diagnostic action			
Enclosure material			

PERFORMANCE CHARACTERISTICS			
Max press at design temp		At	
Min working temperature		Max	
Min material density		Max	
Min ambient working temp		Max	
Contacts ac rating		At max	
Contacts dc rating		At max	

ACCESSORIES	
Mounting hardware	

SPECIAL REQUIREMENTS	
Custom tag	
Reference specification	
Compliance standard	

PHYSICAL DATA			
Estimated weight			
Overall height			
Removal clearance			
Signal conn nominal size		Style	
Mfr reference dwg			

CALIBRATIONS AND TEST		SETPOINT			OUTPUT	
TAG NO/FUNCTIONAL IDENT	MEAS/SIGNAL/TEST	LRV	URV	ACTION	LRV	URV
	Level setpoint-Output					

COMPONENT IDENTIFICATIONS		
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1		RESPONSIBLE ORGANIZATION	LIOUID LEVEL GAGE GLASS w/wo ILLUMINATOR Device Specification	6		SPECIFICATION IDENTIFICATIONS	
2				7		Document no	
3				8		Latest revision	Date
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5				10			

11		CHAMBER AND COVER		60		GAGE VALVES cpntinued	
12		Chamber type		61		Gage/Glass conn type	Style
13		Process conn nominal size	Rating	62		Vent/Drain conn nom size	Rating
14		Process conn termn type	Style	63		Vent/Drain conn type	Style
15		Flange facing finish		64		Max press at design temp	At
16		Process conn location		65		Vessel union/shank type	
17		Vent/Drain conn nom size	Rating	66		Gage union/conn type	
18		Vent/Drain conn type	Style	67		Seat type	
19		Nominal pressure		68		Heating/Cooling type	
20		Number of sections		69		Hand actuator style	
21		Weld pad radius size		70		Stem style	
22		Vent/Drain plug type		71		Body-Bonnet material	
23		Support plates lct		72		Trim material	
24		Heating/Cooling type		73		Ball check shutoff matl	
25		Chamber material		74			
26		Cover material		75			
27		Lining material		76		GAGE ILLUMINATOR	
28		Bolting material		77		Illumination type	
29		Exterior treatment matl		78		Power source	
30		Gasket material		79		Enclosure type no/class	
31		Cushion material		80		Cert/Approval type	
32				81			
33				82		PERFORMANCE CHARACTERISTICS	
34		GAGE GLASS		83		Max press at design temp	At
35		Glass type		84		Min working temperature	Max
36		Glass nominal size		85		Max working pressure	
37		Frost-proof extension lg		86			
38		Visible length		87			
39		Glass/Tube protector type		88		ACCESSORIES	
40		Glass/Tube material		89		Gage scale divisions	
41		Shield material-thickness		90		Gage scale material	
42				91		Vent valves material	
43				92		Drain valves material	
44		INTERCONNECTING ASSEMBLY		93		Bleed fitting material	
45		Interconnection type		94		Flexible insulation	
46		Nipple/Piping nom size	Schedule no	95			
47		Valve offset orientation		96		SPECIAL REQUIREMENTS	
48		Gage orientation		97		Custom tag	
49		Assembled by		98		Reference specification	
50		Nipple/Piping material		99		Compliance standard	
51		Exterior coating material		100		Construction code	
52				101		Cert/Approval type	
53				102			
54		GAGE VALVES		103		PHYSICAL DATA	
55		Valve body type		104		Estimated weight	
56		Bonnet style		105		Gage overall length	
57		Vessel conn nominal size	Rating	106		Min vessel centers	
58		Vessel conn termn type	Style	107		Mfr reference dwg	
59		Gage/Glass conn nom size	Rating	108			

110		CALIBRATIONS AND TEST		INPUT ORTEST			SCALE	
111		TAG NO/FUNCTIONAL IDENT	MEAS/SIGNAL/TEST	LRV	URV		LRV	URV
112			Level -Scale					
113			Test pressure					
114								
115								
116								
117								

118		COMPONENT IDENTIFICATIONS		
119		COMPONENT TYPE	MANUFACTURER	MODEL NUMBER
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RESPONSIBLE ORGANIZATION	MAGNETIC LIQUID LEVEL GAGE OR INDICATOR w/wo SWITCHES Device Specification	SPECIFICATION IDENTIFICATIONS
		Document no
		Latest revision / Date
		Issue status

CAGE OR CHAMBER			
Cage/Chamber type			
Process conn nominal size	Rating		
Process conn termn type	Style		
Lower conn location	Upper		
Top connection style	Bottom		
Vent/Drain conn nom size	Rating		
Vent/Drain termn type	Style		
Cage/Chamber pipe sched			
Cage/Chamber material			
Lining material			
Bolting material			
Flange material			
Gasket material			

SENSING ELEMENT			
Sensor Type			
Sp gr Lower Range-Limit	URL		
Min level span	Max		
Sensor quantity			
Extension length			
Float/Displacer length			
Float/Displacer material			

GAGE OR INDICATOR	
Type	
Seal style	
Visible length	
Gage scale divisions	
Orientation to conn	
Bicolor wafer colors	
Cover material	
Indicator material	

SWITCH MECHANISM			
Housing type			
Element style			
Output signal type			
Enclosure type no/class			
Signal power source			
Contacts arrangement	Quantity		
Signal termination type			
Cert/Approval type			
Enclosure material			

PERFORMANCE CHARACTERISTICS			
Max press at design temp	At		
Min working temperature	Max		
Repeatability			
Min sp gr differential			
Deadband rating			
Min ambient working temp	Max		
Contacts ac rating	At max		
Contacts dc rating	At max		

ACCESSORIES	
Heating kit style	
Isolation valve style	
Vent/Drain valve style	

SPECIAL REQUIREMENTS	
Custom tag	
Reference specification	
Special preparation	
Compliance standard	
Construction code	

PHYSICAL DATA			
Estimated weight			
Overall height			
Removal clearance			
Upper to lower conn lg			
Lower to drain conn lg			
Signal conn nominal size	Style		
Mfr reference dwg			

CALIBRATIONS AND TEST		INPUT OR SETPOINT			SCALE	
TAG NO/FUNCTIONAL IDENT	MEAS/SIGNAL/TEST	LRV	URV	ACTION	LRV	URV
	Level-Scale					
	Level setpoint 1-Output					
	Level setpoint 2-Output					
	Test pressure					

COMPONENT IDENTIFICATIONS		
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RESPONSIBLE ORGANIZATION	PRESSURE GAUGE	SPECIFICATION IDENTIFICATIONS
	Device Specification	Document no
		Latest revision / Date
		Issue status

OPERATING PARAMETERS		DIAL AND POINTER	
Project number	Sub project no	Dial scale type	
Project		Pointer adjustment	
Enterprise		Graduations and color	
Site		Scale range type	
Area	Cell / Unit	Dial material	
Related equipment			
Service			
P&ID/Reference dwg number		SIPHON	
Material name		Siphon type	
Minimum pressure		End conn nominal size	Style
Normal pressure		Overall length	
Maximum pressure		Siphon material	
Normal temperature			
Maximum temperature			
Material phase		PERFORMANCE CHARACTERISTICS	
		Max press at design temp	At
		Min working temperature	Max
PROCESS CONNECTION AND CASE		Min ambient working temp	Max
Case type			
Case style			
Gauge size			
Process conn nominal size	Style		
Process conn location		ACCESSORIES	
Case pressure relief type		Pressure limit valve matl	
Ring style		Restrictor style	
Mounting type		Pressure snubber matl	
Case material		Pulsation dampener matl	
Ring material			
Exterior treatment-color			
Window material		SPECIAL REQUIREMENTS	
Stem material		Custom tag	
Gasket/O ring material		Reference specification	
Liquid fill material		Special preparation	
		Compliance standard	
		Service design	
PRESSURE ELEMENT AND MOVEMENT			
Elastic element type			
Movement style		PHYSICAL DATA	
Nominal accuracy grade		Estimated weight	
Joint type		Maximum thickness	
Element material		Max case outside dia	
Movement material		Mfr reference dwg	

CALIBRATIONS AND TEST			SCALE	
TAG NO/FUNCTIONAL IDENT	PRESSURE OR SCALE		LRV	URV
	Pressure/Scale 1			
	Pressure/Scale 2			

COMPONENT IDENTIFICATIONS

COMPONENT TYPE	MANUFACTURER	MODEL NUMBER

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RESPONSIBLE ORGANIZATION	PRESSURE GAUGE WITH DIAPHRAGM SEAL Device Specification	SPECIFICATION IDENTIFICATIONS
		Document no
		Latest revision / Date
		Issue status

OPERATING PARAMETERS		DIAL AND POINTER continued	
Project number / Sub project no		Graduations and color	
Project		Scale range type	
Enterprise		Dial material	
Site			
Area / Cell / Unit			
Related equipment		DIAPHRAGM SEAL	
Service		Seal type	
		Process conn nom size / Rating	
P&ID/Reference dwg number		Process conn termn type / Style	
Material name		Diaphragm extension lg	
Minimum pressure		Flushing conn quantity	
Normal pressure		Capillary diameter / Length	
Maximum pressure		Instrument conn nom size	
Normal temperature		Diaphragm/wall material	
Maximum temperature		Capillary-armor material	
Material phase		Bolting material	
		Upper housing material	
		Lower housing material	
PROCESS CONNECTION AND CASE		Gasket/O ring material	
Case type		Fill fluid material	
Case style			
Gauge size			
Process conn nominal size / Style		PERFORMANCE CHARACTERISTICS	
Process conn location		Max press at design temp / At	
Case pressure relief type		Min working temperature / Max	
Ring style		Ambient temperature error	
Mounting type		Min ambient working temp / Max	
Case material			
Ring material			
Exterior treatment-color		ACCESSORIES	
Window material		Locking device material	
Stem material			
Seal/O ring material			
Liquid fill material			
		SPECIAL REQUIREMENTS	
		Custom tag	
PRESSURE ELEMENT AND MOVEMENT		Reference specification	
Elastic element type		Special preparation	
Movement style		Compliance standard	
Nominal accuracy grade		Service design	
Joint type			
Element material		PHYSICAL DATA	
Movement material		Estimated weight	
		Maximum thickness	
		Max case outside dia	
DIAL AND POINTER		Mfr reference dwg	
Dial scale type			
Pointer adjustment			

CALIBRATIONS AND TEST			SCALE	
TAG NO/FUNCTIONAL IDENT	PRESSURE OR SCALE		LRV	URV
	Pressure/Scale 1			
	Pressure/Scale 2			

COMPONENT IDENTIFICATIONS		
COMPONENT TYPE	MANUFACTURER	MODEL NUMBER

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RESPONSIBLE ORGANIZATION	DIFFERENTIAL PRESSURE GAUGE	SPECIFICATION IDENTIFICATIONS
	Device Specification	Document no Latest revision / Date Issue status

OPERATING PARAMETERS		DIAL AND POINTER	
Project number	Sub project no	Dial scale type	
Project		Pointer adjustment	
Enterprise		Graduations and color	
Site		Scale range type	
Area	Cell / Unit	Dial material	
Related equipment			
Service			
P&ID/Reference dwg number		SIPHON	
Material name		Siphon type	
Maximum pressure		End conn nominal size	Style
Maximum temperature		Overall length	
Material phase		Siphon material	
Min diff pressure			
Norm diff pressure			
Max diff pressure		PERFORMANCE CHARACTERISTICS	
		Max press at design temp	At
		Min working temperature	Max
PROCESS CONNECTION AND CASE		Max diff pressure	
Case type		Min ambient working temp	Max
Case style			
Gauge size			
Process conn nominal size	Style		
Process conn location		ACCESSORIES	
Case pressure relief type		Pressure limit valve matl	
Ring style		Restrictor style	
Mounting type		Pressure snubber matl	
Case material		Pulsation dampener matl	
Ring material		Scale overlays	
Exterior treatment-color			
Window material		SPECIAL REQUIREMENTS	
Stem material		Custom tag	
Gasket/O ring material		Reference specification	
Liquid fill material		Special preparation	
		Compliance standard	
		Service design	
PRESSURE ELEMENT AND MOVEMENT			
Elastic element type			
Movement style		PHYSICAL DATA	
Nominal accuracy grade		Estimated weight	
Joint type		Maximum thickness	
Element material		Max case outside dia	
Movement material		Mfr reference dwg	

CALIBRATIONS AND TEST			SCALE	
TAG NO/FUNCTIONAL IDENT	PRESSURE OR SCALE		LRV	URV
	Diff pressure/Scale 1			
	Diff pressure/Scale 2			

COMPONENT IDENTIFICATIONS		
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RESPONSIBLE ORGANIZATION	PRESSURE TRANSMITTER	SPECIFICATION IDENTIFICATIONS	
	Device Specification	Document no	
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TRANSMITTER BODY	
Body/Flange type	
Process conn nominal size	Rating
Process conn termn type	Style
Vent/Drain location	
Mounting type	
Body/Flange material	
Vent/Drain material	
Bolting material	
Flange adapter material	
Gasket/O ring material	
Mounting kit material	

SENSING ELEMENT	
Detector type	
Min pressure span	Max
Diaphragm/Wetted material	
Fill fluid material	

TRANSMITTER	
Output signal type	
Enclosure type no/class	
Characteristic curve	
Digital communication std	
Signal power source	
Transient protection	
Integral indicator style	
Signal termination type	
Cert/Approval type	
Span-Zero adjust lct	
Failure/Diagnostic action	
Enclosure material	

PERFORMANCE CHARACTERISTICS			
Max press at design temp		At	
Min working temperature		Max	
Accuracy rating			
Pressure LRL		URL	
Min ambient working temp		Max	

ACCESSORIES	
Air set filter style	
Air set gauges	
Heating kit style	
Remote indicator style	
Manifold valve style	

SPECIAL REQUIREMENTS	
Custom tag	
Reference specification	
Special preparation	
Compliance standard	
Software configuration	

PHYSICAL DATA			
Estimated weight			
Overall height			
Removal clearance			
Signal conn nominal		Style	
Mfr reference dwg			

CALIBRATIONS AND TEST		INPUT			OUTPUT OR SCALE	
TAG NO/FUNCTIONAL IDENT	MEAS/SIGNAL/TEST	LRV	URV	ACTION	LRV	URV
	Pressure-Analog output					
	Pressure-Scale					
	Pressure-Digital output					
	Temp-Digital output					

COMPONENT IDENTIFICATIONS		
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RESPONSIBLE ORGANIZATION	PRESSURE TRANSMITTER WITH DIAPHRAGM SEAL Device Specification	SPECIFICATION IDENTIFICATIONS
		Document no
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TRANSMITTER BODY			
Body/Flange type			
Process conn nominal size		Rating	
Process conn termn type		Style	
Vent/Drain location			
Mounting type			
Body/Flange material			
Vent/Drain material			
Bolting material			
Flange adapter material			
Gasket/O ring material			
Mounting kit material			

SENSING ELEMENT			
Detector type			
Min pressure span		Max	
Diaphragm/Wetted material			
Fill fluid material			

TRANSMITTER			
Output signal type			
Enclosure type no/class			
Characteristic curve			
Digital communication std			
Signal power source			
Transient protection			
Integral indicator style			
Signal termination type			
Cert/Approval type			
Span-Zero adjust lct			
Failure/Diagnostic action			
Enclosure material			

DIAPHRAGM SEAL			
Seal type			
Process conn nominal size		Rating	
Process conn termn type		Style	
Diaphragm extension lg			
Flushing conn quantity			
Instrument conn nom size			
Capillary/Fitting dia		Length	
Diaphragm material			
Capillary-armor matl			
Bolting material			
Upper housing material			
Lower housing/Flange matl			
Gasket/O ring material			
Fill fluid material			

PERFORMANCE CHARACTERISTICS			
Max press at design temp		At	
Min working temperature		Max	
Accuracy rating			
Pressure Lower Range-Limit		URL	
Ambient temperature error			
Min ambient working temp		Max	

ACCESSORIES	
Air set filter style	
Air set gauges	
Heating kit style	
Remote indicator style	

SPECIAL REQUIREMENTS	
Custom tag	
Reference specification	
Special preparation	
Compliance standard	
Software configuration	

PHYSICAL DATA			
Estimated weight			
Overall height			
Removal clearance			
Signal conn nominal size		Style	
Mfr reference dwg			

CALIBRATIONS AND TEST		INPUT			OUTPUT OR SCALE	
TAG NO/FUNCTIONAL IDENT	MEAS/SIGNAL/TEST	LRV	URV	ACTION	LRV	URV
	Pressure-Analog output					
	Pressure-Scale					
	Pressure-Digital output					
	Temp-Digital output					

COMPONENT IDENTIFICATIONS		
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TRANSMITTER BODY			
Body/Flange type			
Process conn nominal size		Rating	
Process conn termn type		Style	
Vent/Drain location			
Mounting type			
Body/Flange material			
Vent/Drain material			
Bolting material			
Flange adapter material			
Gasket/O ring material			
Mounting kit material			

SENSING ELEMENT			
Detector type			
Min diff press span		Max	
Diaphragm/Wetted material			
Fill fluid material			

TRANSMITTER	
Output signal type	
Enclosure type no/class	
Characteristic curve	
Digital communication std	
Signal power source	
Transient protection	
Integral indicator style	
Signal termination type	
Cert/Approval type	
Span-Zero adjust lct	
Failure/Diagnostic action	
Enclosure material	

PERFORMANCE CHARACTERISTICS			
Max press at design temp		At	
Min working temperature		Max	
Accuracy rating			
Diff pressure LRL		URL	
Min ambient working temp		Max	

ACCESSORIES	
Air set filter style	
Air set gauges	
Heating kit style	
Remote indicator style	
Manifold valve style	

SPECIAL REQUIREMENTS	
Custom tag	
Reference specification	
Special preparation	
Compliance standard	
Software configuration	

PHYSICAL DATA			
Estimated weight			
Overall height			
Removal clearance			
Signal con nominal size		Style	
Mfr reference dwg			

CALIBRATIONS AND TEST		INPUT			OUTPUT OR SCALE	
TAG NO/FUNCTIONAL IDENT	MEAS/SIGNAL/TEST	LRV	URV	ACTION	LRV	URV
	Diff press-Analog output					
	Diff pressure-Scale					
	Diff press-Digital output					
	Press-Digital output					
	Temp-Digital output					

COMPONENT IDENTIFICATIONS		
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RESPONSIBLE ORGANIZATION	DIFFERENTIAL PRESSURE TRANSMITTER WITH DIAPHRAGM SEALS Device Specification	SPECIFICATION IDENTIFICATIONS	
		Document no	
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TRANSMITTER BODY			
Body/Flange type			
Process conn nominal size		Rating	
Process conn termn type		Style	
Vent/Drain location			
Mounting type			
Body/Flange material			
Vent/Drain material			
Bolting material			
Flange adapter material			
Gasket/O ring material			
Mounting kit material			

SENSING ELEMENT			
Detector type			
Min diff pressure span		Max	
Diaphragm/Wetted material			
Fill fluid material			

TRANSMITTER			
Output signal type			
Enclosure type no/class			
Characteristic curve			
Digital communication std			
Signal power source			
Transient protection			
Integral indicator style			
Signal termination type			
Cert/Approval type			
Span-Zero adjust lct			
Failure/Diagnostic action			
Enclosure material			

HIGH SIDE SEAL			
Seal type			
Process conn nominal size		Rating	
Process conn termn type		Style	
Diaphragm extension lg			
Flushing conn quantity			
Instrument conn nom size			
Capillary/Fitting dia		Length	
Diaphragm material			
Capillary-armor matl			
Bolting material			
Upper housing material			
Lower housing/Flange matl			
Gasket/O ring material			
Fill fluid material			

LOW SIDE SEAL			
Seal type			
Process conn nominal size		Rating	
Process conn termn type		Style	
Diaphragm extension lg			
Flushing conn quantity			
Instrument conn nom size			
Capillary/Fitting dia		Length	
Diaphragm material			
Capillary-armor material			
Bolting material			
Upper housing material			
Lower housing/Flange matl			
Gasket/O ring material			
Fill fluid material			

PERFORMANCE CHARACTERISTICS			
Max press at design temp		At	
Min working temperature		Max	
Accuracy rating			
Diff pressure LRL		URL	
Fill fluid sp gr		At temp	
Ambient temperature error			
Min ambient working temp		Max	

ACCESSORIES	
Air set filter style	
Air set gauges	
Heating kit style	
Remote indicator style	

SPECIAL REQUIREMENTS	
Custom tag	
Reference specification	
Special preparation	
Compliance standard	
Software configuration	

PHYSICAL DATA			
Estimated weight			
Overall height			
Removal clearance			
Signal conn nominal size		Style	
Mfr reference dwg			

CALIBRATIONS AND TEST		INPUT			OUTPUT OR SCALE	
TAG NO/FUNCTIONAL IDENT	MEAS/SIGNAL/TEST	LRV	URV	ACTION	LRV	URV
	Diff press-Analog output					
	Diff press-Scale					
	Diff press-Digital output					
	Press-Digital output					
	Temp-Digital output					

COMPONENT IDENTIFICATIONS		
COMPONENT TYPE	MANUFACTURER	MODEL NUMBER

Rev	Date	Revision Description	By	Appv1	Appv2	Appv3	REMARKS

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RESPONSIBLE ORGANIZATION	PRESSURE SWITCH w/wo TRANSMITTER Device Specification	SPECIFICATION IDENTIFICATIONS	
		Document no	
		Latest revision	Date
		Issue status	

SWITCH BODY	
Body type	
Process conn nominal size	Rating
Process conn termn type	Style
Body material	
Seal/O ring material	

SENSING ELEMENT	
Sensor element type	
Adjustable LRL	URL
Diaphragm/Wetted material	

SWITCH MECHANISM w/wo TRANSMITTER	
Housing type	
Element Style	
Output signal type	
Enclosure type no/class	
Reset style	
Set point Adjustment type	
Signal power source	
Measurement type	
Contacts arrangement	Quantity
Failsafe style	
Integral indicator style	
Signal termination type	
Cert/Approval type	
Mounting type	
Dead band type	
Enclosure material	
Exterior treatment matl	

PERFORMANCE CHARACTERISTICS	
Max press at design temp	At
Min working temperature	Max
Output accuracy rating	
Repeatability	
Press Lower Range-limit	URL
Max overrange limit	
Dead band rating	
Min ambient working temp	Max
Contacts ac rating	At max
Contacts dc rating	At max

ACCESSORIES	
Sealed leads adapter	
Breather/Drain style	

SPECIAL REQUIREMENTS	
Custom tag	
Reference specification	
Special preparation	
Compliance standard	

PHYSICAL DATA	
Estimated weight	
Overall height	
Removal clearance	
Signal conn nominal size	Style
Mfr reference dwg	

CALIBRATIONS AND TEST		INPUT/SETPOINT/TEST			OUTPUT OR SCALE	
TAG NO/FUNCTIONAL IDENT	MEAS/SIGNAL/TEST	LRV	URV	ACTION	LRV	URV
	Press setpoint 1-Output					
	Press setpoint 2-Output					
	Pressure-Analog output					
	Pressure-Scale					

COMPONENT IDENTIFICATIONS		
COMPONENT TYPE	MANUFACTURER	MODEL NUMBER

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RESPONSIBLE ORGANIZATION	DIFFERENTIAL PRESSURE SWITCH w/wo TRANSMITTER Device Specification	SPECIFICATION IDENTIFICATIONS	
		Document no	
		Latest revision	Date
		Issue status	

SWITCH BODY	
Body type	
Process conn nominal size	Rating
Process conn termn type	Style
Body material	
Seal/O ring material	

SENSING ELEMENT	
Sensor element type	
Adjustable LRL	URL
Diaphragm/Wetted material	

SWITCH MECHANISM w/wo TRANSMITTER	
Housing type	
Element Style	
Output signal type	
Enclosure type no/class	
Reset style	
Set point Adjustment type	
Signal power source	
Measurement type	
Contacts arrangement	Quantity
Failsafe style	
Integral indicator style	
Signal termination type	
Cert/Approval type	
Mounting type	
Dead band type	
Enclosure material	
Exterior coating material	

PERFORMANCE CHARACTERISTICS	
Max press at design temp	At
Min working temperature	Max
Output accuracy rating	
Repeatability	
Diff press Lower Range-Limit	URL
Max overrange limit	
Dead band rating	
Min ambient working temp	Max
Contacts ac rating	At max
Contacts dc rating	At max

ACCESSORIES	
Sealed leads adapter	
Breather/Drain style	

SPECIAL REQUIREMENTS	
Custom tag	
Reference specification	
Special preparation	
Compliance standard	

PHYSICAL DATA	
Estimated weight	
Overall height	
Removal clearance	
Signal conn nominal size	Style
Mfr reference dwg	

CALIBRATIONS AND TEST		INPUT OR SETPOINT			OUTPUT OR SCALE	
TAG NO/FUNCTIONAL IDENT	MEAS/SIGNAL/TEST	LRV	URV	ACTION	LRV	URV
	DP setpoint 1-Output					
	DP setpoint 2-Output					
	Diff press-Analog output					
	Diff pressure-Scale					

COMPONENT IDENTIFICATIONS		
COMPONENT TYPE	MANUFACTURER	MODEL NUMBER

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RESPONSIBLE ORGANIZATION	RECEIVER GAUGE	SPECIFICATION IDENTIFICATIONS	
	Device Specification	Document no	
		Latest revision	Date
		Issue status	

SIGNAL CONNECTION AND CASE			
Case type			
Gauge style			
Gauge size			
Signal conn nominal size		Style	
Signal conn location			
Case pressure relief type			
Ring style			
Mounting type			
Case material			
Ring material			
Exterior treatment-color			
Window material			
Stem material			
Gasket/O ring material			

PRESSURE ELEMENT AND MOVEMENT	
Elastic element type	
Movement style	
Nominal accuracy grade	
Joint type	
Internal stops	
Element material	
Movement material	

DIAL AND POINTER	
Dial scale type	
Pointer adjustment	
Graduations and color	
Scale range type	
Dial material	

PERFORMANCE CHARACTERISTICS			
Min ambient working temp		Max	

ACCESSORIES	
Mounting hardware	

SPECIAL REQUIREMENTS	
Custom tag	
Reference specification	
Compliance standard	

PHYSICAL DATA	
Estimated weight	
Maximum thickness	
Max case outside dia	
Mfr reference dwg	

CALIBRATIONS AND TEST		INPUT			SCALE	
TAG NO/FUNCTIONAL IDENT	PRESSURE OR SCALE	LRV	URV		LRV	URV
	Input signal-Scale 1					
	Input signal-Scale 2					

COMPONENT IDENTIFICATIONS		
COMPONENT TYPE	MANUFACTURER	MODEL NUMBER

Rev	Date	Revision Description	By	Appv1	Appv2	Appv3	REMARKS

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RESPONSIBLE ORGANIZATION	BIMETALLIC THERMOMETER w/wo THERMOWELL Device Specification	SPECIFICATION IDENTIFICATIONS
		Document no
		Latest revision / Date
		Issue status

OPERATING PARAMETERS		THERMOWELL OR PROTECTING TUBE	
Project number	Sub project no	Construction type	
Project		Shank style	
Enterprise		Process conn nominal size	Rating
Site		Process conn termn type	Style
Area	Cell / Unit	Internal conn nom size	Style
Related equipment		Bore diameter	
Service		Outside dia at support	
		Outside diameter at tip	
P&ID/Reference dwg number		Insertion length "U"	
Material name		Lagging extension lg "T"	
Maximum pressure		Thermowell/Tube material	
Minimum temperature		Sheath material-thickness	
Normal temperature			
Maximum temperature			
Material phase			
Maximum fluid velocity			

PERFORMANCE CHARACTERISTICS	
Max press at design temp	At
Min working temperature	Max
Max fluid velocity limit	At temp
Sensitive portion length	
Min ambient working temp	Max

PROCESS CONNECTION AND CASE		ACCESSORIES	
Case type		Bushing nominal size	
Case style		Bushing material	
Case size			
Connector conn nom size			
Connector conn type	Style		
Connection location			
Ring style			
Stem outside diameter	Length		
Stem/Bulb material			
Case material			
Ring material			
Exterior treatment-color			
Window material			
Connector material			

SENSING ELEMENT		SPECIAL REQUIREMENTS	
Element type		Custom tag	
Nominal accuracy grade		Reference specification	
		Compliance standard	
		Construction code	
		Calculation report	

DIAL AND POINTER		PHYSICAL DATA	
Dial scale type		Estimated weight	
Pointer adjustment		Removal clearance	
Graduations and color		Maximum thickness	
		Max case outside dia	
		Mfr reference dwg	

CALIBRATIONS AND TEST		TEST			SCALE	
TAG NO/FUNCTIONAL IDENT	MEAS OR TEST		URV		LRV	URV
	Temperature/Scale 1					
	Temperature/Scale 2					
	Test pressure					

COMPONENT IDENTIFICATIONS		
COMPONENT TYPE	MANUFACTURER	MODEL NUMBER

Rev	Date	Revision Description	By	Appv1	Appv2	Appv3	REMARKS

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		Document no
		Latest revision / Date
		Issue status

OPERATING PARAMETERS		BULB AND CASE		CAPILLARY AND SENSING ELEMENT		DIAL AND POINTER	
Project number	Sub project no	Case type		Thermal system type		Dial scale type	
Project		Stem/Bulb extension style		Compensation style		Pointer adjustment	
Enterprise		Case size		Nominal accuracy grade			
Site		Connector conn nom size		Capillary length			
Area	Cell / Unit	Connector conn type	Style	Capillary material			
Related equipment		Connection location		Coating-armor material			
Service		Ring style					
		Stem/Bulb outside dia	Length				
P&ID/Reference dwg number		Mounting type					
Material name		Stem/Bulb material					
Maximum pressure		Case-Ring material					
Minimum temperature		Exterior treatment-color					
Normal temperature		Window material					
Maximum temperature		Connector material					
Material phase							
Maximum fluid velocity							

THERMOWELL/PROTECTING TUBE continued		THERMOWELL OR PROTECTING TUBE		PERFORMANCE CHARACTERISTICS		ACCESSORIES	
Graduations and color		Construction type		Max press at design temp	At	Bushing nominal size	
		Shank style		Min working temperature	Max	Bushing material	
		Process conn nominal size	Rating	Max fluid velocity limit	At temp		
		Process conn termn type	Style	Sensitive portion length			
		Internal conn nom size	Style	Ambient temperature error			
		Bore diameter		Min ambient working temp	Max		
		Outside dia at support					
		Outside diameter at tip					
		Insertion length “U”					
		Lagging extension lg “T”					
		Thermowell/Tube material					
		Sheath material-thickness					

SPECIAL REQUIREMENTS		PHYSICAL DATA	
Custom tag		Estimated weight	
Reference specification		Removal clearance	
Compliance standard		Maximum thickness	
Construction code		Max case outside dia	
Calculation report		Mfr reference dwg	

CALIBRATIONS AND TEST		TEST			SCALE	
TAG NO/FUNCTIONAL IDENT	TEMP OR TEST		URV		LRV	URV
	Temperature/Scale 1					
	Temperature/Scale 2					
	Test pressure					

COMPONENT IDENTIFICATIONS

COMPONENT TYPE	MANUFACTURER	MODEL NUMBER

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PROCESS CONNECTION AND BULB			
Bulb type			
Stem/Bulb extension style			
Connector conn nom size			
Connector conn type		Style	
Fitting type			
Stem/Bulb diameter		Length	
Bulb extension length			
Stem/Bulb material			
Connector material			

CAPILLARY AND SENSING ELEMENT			
Thermal system type			
Compensation style			
Temperature LRL		URL	
Capillary length			
Capillary material			
Coating-armor material			

SWITCH MECHANISM			
Type			
Element style			
Enclosure type no/class			
Reset style			
Set point adjustment lct			
Signal power source			
Dead band type			
Contacts arrangement		Quantity	
Integral indicator style			
Signal termination type			
Cert/Approval type			
Mounting type			
Signal conn orientation			
Enclosure material			
Exterior coating material			

THERMOWELL OR PROTECTING TUBE			
Construction type			
Shank style			
Process conn nominal size		Rating	
Process conn termn type		Style	
Internal conn nom size		Style	
Bore diameter			
Outside dia at support			
Outside dia at tip			
Insertion length “U”			
Lagging extension lg “T”			
Thermowell/Tube material			
Sheath material-thickness			

PERFORMANCE CHARACTERISTICS			
Max press at design temp		At	
Min working temperature		Max	
Max fluid velocity limit		At temp	
Repeatability			
Max overrange limit			
Ambient temperature error			
Dead band rating			
Min ambient working temp		Max	
Contacts ac rating		At max	
Contacts dc rating		At max	

ACCESSORIES	
Bushing nominal size	
Bushing material	
Sealed leads adapter	
Breather/Drain style	

SPECIAL REQUIREMENTS	
Custom tag	
Reference specification	
Special preparation	
Compliance standard	
Calculation report	

PHYSICAL DATA			
Estimated weight			
Overall height			
Removal clearance			
Signal conn nominal size		Style	
Mfr reference dwg			

CALIBRATIONS AND TEST		INPUT/SETPOINT/TEST			OUTPUT OR SCALE	
TAG NO/FUNCTIONAL IDENT	MEAS OR TEST	LRV	URV	ACTION	LRV	URV
	Temp setpoint 1-Output					
	Temp setpoint 2-Output					
	Temperature-Scale					
	Test pressure					

COMPONENT IDENTIFICATIONS		
COMPONENT TYPE	MANUFACTURER	MODEL NUMBER

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		Document no	
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PROCESS CONNECTION AND BULB			
Bulb type			
Stem/Bulb extension style			
Connector conn nom size			
Connector conn type		Style	
Fitting type			
Stem/Bulb diameter		Length	
Bulb extension length			
Stem/Bulb material			
Connector material			
Packing material			

CAPILLARY AND SENSING ELEMENT			
Thermal system type			
Compensation style			
Temperature LRL		URL	
Min temperature span		Max	
Capillary length			
Capillary material			
Coating-armor material			

TRANSMITTER	
Housing type	
Output signal type	
Enclosure type no/class	
Signal power source	
Integral indicator style	
Signal termination type	
Cert/Approval type	
Mounting type	
Enclosure material	

THERMOWELL OR PROTECTING TUBE			
Construction type			
Shank style			
Process conn nominal size		Rating	
Process conn termn type		Style	
Internal conn nom size		Style	
Bore diameter			
Outside dia at support			
Outside dia at tip			
Insertion length “U”			
Lagging extension lg “T”			
Thermowell/tube material			
Sheath material-thickness:			

PERFORMANCE CHARACTERISTICS			
Max press at design temp		At	
Min working temperature		Max	
Max fluid velocity limit		At temp	
Accuracy rating			
Max overrange limit			
Ambient temperature error			
Min ambient working temp		Max	

ACCESSORIES	
Bushing nominal size	
Bushing material	
Air set filter style	
Pressure gauges	

SPECIAL REQUIREMENTS	
Custom tag	
Reference specification	
Compliance standard	
Construction code	
Calculation report	

PHYSICAL DATA			
Estimated weight			
Overall height			
Removal clearance			
Signal conn nominal size		Style	
Mfr reference dwg			

CALIBRATIONS AND TEST		INPUT OR TEST			OUTPUT OR SCALE	
TAG NO/FUNCTIONAL IDENT	MEAS OR TEST	LRV	URV	ACTION	LRV	URV
	Temp-Analog output					
	Temperature-Scale					
	Test pressure					

COMPONENT IDENTIFICATIONS		
COMPONENT TYPE	MANUFACTURER	MODEL NUMBER

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RESPONSIBLE ORGANIZATION	RTD ASSEMBLY w/wo THERMOWELL Device Specification	SPECIFICATION IDENTIFICATIONS	
		Document no	
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PROTECTIVE SHEATH AND FITTING			
Housing type			
Pad/Collar type			
Fitting conn nominal size		Style	
Mounting fitting type			
Sheath alignment			
Sheath outside diameter		Length	
Spring loading			
Sheath material			
Fitting material			

SENSING ELEMENT			
Sensor type			
Sensor quantity			
Temperature coefficient			
Tolerance class			
Nominal resistance		At temp	
Configuration-no of wires			
Sensor material			
Insulator material			

LEAD WIRE AND HEAD EXTENSION			
Extension type			
Ext wire nom size-type			
Extension/Lead length			
Nipple-union nom size		Rating	
Nipple pipe schedule no			
Transition type			
Termination style			
Connecting wire length			
Shield - ground wire			
Nipple material			
Union/Coupling material			
Coating-armor material			
Extension wire material			
Ext wire insulation matl			

CONNECTION HEAD			
Housing type			
Cover style			
Element conn nominal size		Style	
Signal conn nominal size		Style	
Enclosure type no/class			
Grounding terminal lct			
Enclosure material			
Terminal block material			
Terminal material			

THERMOWELL OR PROTECTING TUBE			
Construction type			
Shank style			
Process conn nominal size		Rating	
Process conn termn type		Style	
Internal conn nom size		Style	
Bore diameter			
Outside dia at support			
Outside dia at tip			
Insertion length “U”			
Lagging extension lg “T”			
Thermowell/Tube material			
Sheath material-thickness			

PERFORMANCE CHARACTERISTICS			
Max press at design temp		At	
Min working temperature		Max	
Max fluid velocity limit		At temp	
Temp Lower Range-Limit		URL	
Min insertion length			
Min ambient working temp		Max	

ACCESSORIES	
Moisture seal style	
Bayonet adapter size	

SPECIAL REQUIREMENTS	
Custom tag	
Reference specification	
Compliance standard	
Construction code	
Calculation report	
Calibration report	

PHYSICAL DATA	
Estimated weight	
Overall length	
Removal clearance	
Mfr reference dwg	

CALIBRATIONS AND TEST		INPUT OR TEST			OUTPUT	
TAG NO/FUNCTIONAL IDENT	MEAS/SIGNAL/TEST	LRV	URV		LRV	URV
	Temp-Output signal					
	Test pressure					

COMPONENT IDENTIFICATIONS		
COMPONENT TYPE	MANUFACTURER	MODEL NUMBER

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RESPONSIBLE ORGANIZATION	RTD/THERMOCOUPLE TEMPERATURE TRANSMITTER OR SWITCH Device Specification	SPECIFICATION IDENTIFICATIONS	
		Document no	
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TRANSMITTER OR SWITCH			
Housing type			
Input sensor type			
Output signal type			
Min measurement span		Max	
Temp coef/Tolerance cl			
Isolation type			
Enclosure type no/class			
Adjustment type			
Characteristic curve			
Digital communication std			
Signal power source			
Measurement type			
Configuration-no of wires			
Contacts arrangement		Quantity	
Failsafe style			
Transient protection			
Integral indicator style			
Signal termination type			
Cert/Approval type			
Mounting type			
Failure/Diagnostic action			
Dead band type			
Switch time delay			
Temp compensation type			
Enclosure material			
Mounting kit material			

PERFORMANCE CHARACTERISTICS			
Accuracy rating			
Measurement LRL		URL	
Min ambient working temp		Max	
Contacts ac rating		At max	
Contacts dc rating		At max	

ACCESSORIES	
Remote indicator style	
Indicator enclosure	
Air set filter style	
Air set gauges	

SPECIAL REQUIREMENTS	
Custom tag	
Reference specification	
Compliance standard	
Calibration report	
Software configuration	

PHYSICAL DATA			
Estimated weight			
Overall height			
Removal clearance			
Signal conn nominal size		Style	
Mfr reference dwg			

CALIBRATIONS AND TEST		INPUT OR SETPOINT			OUTPUT OR SCALE	
TAG NO/FUNCTIONAL IDENT	MEAS/SIGNAL/TEST	LRV	URV	ACTION	LRV	URV
	Temp- Analog output 1					
	Temp-Analog output 2					
	Temp-Scale					
	Temp diff-Digital output					
	Temp-Digital output					
	Termn temp-Digital output					
	Temp setpoint 1-Output					
	Temp setpoint 2-Output					
	Temp setpoint 3-Output					
	Temp setpoint 4-Output					

COMPONENT IDENTIFICATIONS		
COMPONENT TYPE	MANUFACTURER	MODEL NUMBER

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RESPONSIBLE ORGANIZATION	THERMOCOUPLE ASSEMBLY w/wo THERMOWELL Device Specification	SPECIFICATION IDENTIFICATIONS
		Document no Latest revision / Date Issue status

PROTECTIVE SHEATH AND FITTING			
Housing type			
Pad/Collar type			
Fitting conn nominal size		Style	
Mounting fitting type			
Sheath alignment			
Sheath outside diameter		Length	
Spring loading			
Sheath/Braid material			
Fitting material			

SENSING ELEMENT			
Sensor type		Quantity	
Wire nominal size			
Thermocouple type			
Tolerance class			
Measuring junction			
Thermocouple wire matl			
Insulator material			

LEAD WIRE AND HEAD EXTENSION			
Extension type			
Ext wire nom size-type			
Extension/Lead length			
Nipple-union nom size		Rating	
Nipple pipe sched no			
Transition type			
Termination style			
Connecting wire length			
Shield - ground wire			
Nipple material			
Union/Coupling material			
Coating-armor material			
Extension wire material			
Ext wire insulation matl			

CONNECTION HEAD			
Housing type			
Cover style			
Element conn nominal size		Style	
Signal conn nominal size		Style	
Enclosure type no/class			
Grounding terminal lct			
Enclosure material			
Terminal block material			
Terminal material			

THERMOWELL OR PROTECTING TUBE			
Construction type			
Shank style			
Process conn nominal size		Rating	
Process conn termn type		Style	
Internal conn nom size		Style	
Bore diameter			
Outside dia at support			
Outside dia at tip			
Insertion length “U”			
Lagging extension lg “T”			
Thermowell/Tube material			
Sheath material-thickness			

PERFORMANCE CHARACTERISTICS			
Max press at design temp		At	
Min working temperature		Max	
Max fluid velocity limit		At temp	
Temp Lower Range-Limit		URL	
Min ambient working temp		Max	

ACCESSORIES	
Moisture seal style	
Bayonet adapter size	

SPECIAL REQUIREMENTS	
Custom tag	
Reference specification	
Compliance standard	
Construction code	
Calculation report	
Calibration report	

PHYSICAL DATA	
Estimated weight	
Overall length	
Removal clearance	
Mfr reference dwg	

CALIBRATIONS AND TEST		INPUT OR TEST			OUTPUT	
TAG NO/FUNCTIONAL IDENT	MEAS/SIGNAL/TEST	LRV	URV		LRV	URV
	Temp-Output signal					
	Test pressure					

COMPONENT IDENTIFICATIONS		
COMPONENT TYPE	MANUFACTURER	

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RESPONSIBLE ORGANIZATION	THERMAL RADIATION TEMPERATURE SENSOR w/wo MONITOR Device Specification	SPECIFICATION IDENTIFICATIONS
		Document no
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SENSING ELEMENT	
Sensor type	
Temperature LRL / URL	
Spectral response	
Focus type	

HEAD AND VIEWER	
Head type	
Housing style	
Enclosure type no/class	
Sighting type	
Lens purge style	
Conn adapter type	
Cable length	
Signal termination type	
Cert/Approval type	
Mounting type	
Enclosure material	

TRANSMITTER OR MULTIPLEXER	
Type	
Signal selector type	
Cable length	
Output signal type	
Enclosure type no/class	
Signal power source	
Adjustment type	
Number input sensors	

MONITOR	
Housing type	
External input type	
Output signal type	
Zero-span adjustment lct	
Enclosure type no/class	
Measurement type	
Digital communication std	
Signal power source	
Dead band type	
Contacts arrangement / Quantity	
Emissivity adjustment	
Failsafe style	
Integral indicator style	
Signal termination type	
Cert/Approval type	
Mounting type	
Temperature compensation	

MONITOR Continued	
Failure/Diagnostic action	
Enclosure material	

PERFORMANCE CHARACTERISTICS	
Min head working temp / Max	
Temp accuracy rating	
Focus point D:S ratio	
Far point D:S ratio	
Min monitor amb working / Max	
Contacts ac rating / At max	
Contacts dc rating / At max	

ACCESSORIES	
Blast gate style	
Cooling flow regulator	
Sight tube material	
Printer	

SPECIAL REQUIREMENTS	
Custom tag	
Reference specification	
Compliance standard	
Calibration report	
Software configuration	

PHYSICAL DATA	
Estimated weight	
Overall height	
Removal clearance	
Signal conn nominal size / Style	
Mfr reference dwg	

CALIBRATIONS AND TEST		INPUT OR SETPOINT			OUTPUT OR SCALE	
TAG NO/FUNCTIONAL IDENT	MEAS/SIGNAL/SCALE	LRV	URV	ACTION	LRV	URV
	Temp-Analog output 1					
	Temp-Analog output 2					
	Temp-Analog output 3					
	Temperature-Scale					
	Temp setpoint 1-Output					
	Temp setpoint 2-Output					

COMPONENT IDENTIFICATIONS		
COMPONENT TYPE	MANUFACTURER	MODEL NUMBER

Rev	Date	Revision Description	By	Appv1	Appv2	Appv3	REMARKS

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		Document no
		Latest revision / Date
		Issue status

OPERATING PARAMETERS			
Project number		Sub project no	
Project			
Enterprise			
Site			
Area		Cell / Unit	
Related equipment			
Service			
P&ID/Reference dwg number			
Material name			
Maximum pressure			
Minimum temperature			
Normal temperature			
Maximum temperature			
Material phase			
Maximum fluid velocity			

THERMOWELL OR PROTECTING TUBE			
Construction type			
Shank style			
Process conn nominal size		Rating	
Process conn termn type		Style	
Internal conn nom size		Style	
Cap/Plug style			
Bore diameter			
Outside dia at support			
Outside diameter at tip			
Insertion length “U”			
Lagging extension lg “T”			
Thermowell/Tube material			
Sheath material-thickness			
Cap/Plug material			

EXTENSION ASSEMBLY			
Extension type			
Extension length			
Nipple-union nom size		Rating	
Nipple pipe schedule no			
Nipple material			
Union/Coupling material			

CONNECTION HEAD	
Housing type	
Cover style	

CONNECTION HEAD continued			
Element conn nominal size		Style	
Signal conn nominal size		Style	
Enclosure type no/class			
Grounding terminal lct			
Enclosure material			
Terminal block material			
Terminal material			

PERFORMANCE CHARACTERISTICS			
Max press at design temp		At	
Min working temperature		Max	
Max fluid velocity limit		At temp	

ACCESSORIES	

SPECIAL REQUIREMENTS	
Custom tag	
Reference specification	
Compliance standard	
Construction code	
Calculation report	

PHYSICAL DATA	
Estimated weight	
Overall length	
Removal clearance	
Mfr reference dwg	

CALIBRATIONS AND TEST		TEST	
TAG NO/FUNCTIONAL IDENT	TEST		URV
	Test pressure		

COMPONENT IDENTIFICATIONS		
COMPONENT TYPE	MANUFACTURER	MODEL NUMBER

Rev	Date	Revision Description	By	Appv1	Appv2	Appv3	REMARKS

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		Issue status	

These general or special requirements shall be considered as part of the requirements for the equipment covered by the individual requisition and/or specification sheets.

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Requirements	Page

Rev	Date	Revision Description	By	Appv1	Appv2	Appv3	REMARKS

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Annex C (informative) – Specification forms pick list and contextual help references

This annex is included for informational purposes and is not part of this Technical Report.

Due to the large size of the pick list files for all of the approved forms, they are not attached to this Technical Report. The latest versions of the pick lists may be viewed and downloaded for free from the ISA website at www.isa.org/standards.

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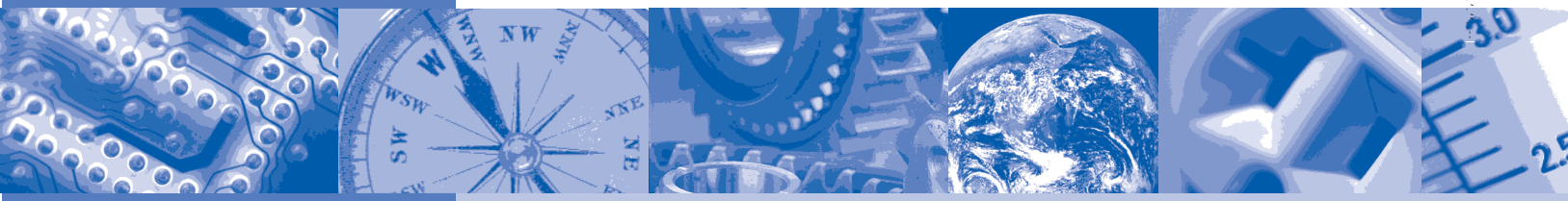
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STANDARD

ISA–20.00.03–2001



Specification Forms for Process Measurement and Control Instruments Part 3: Form Requirements and Development Guidelines

Approved 04 April 2001

ISA–The Instrumentation, Systems, and Automation Society

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ISA–20.00.03–2001
Specification Forms for Process Measurement and Control Instruments
Part 3: Form Requirements and Development Guidelines

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Preface

This preface, as well as all footnotes and annexes, is included for information purposes and is not part of ISA–20.00.03–2001.

The standards referenced within this document may contain provisions that, through reference in this text, constitute requirements of this document. At the time of publication, the editions indicated were valid. All standards are subject to revision, and parties to agreements based on this document are encouraged to investigate the possibility of applying the most recent editions of the standards indicated within this document. Members of IEC and ISO maintain registers of currently valid International Standards. ANSI maintain registers of currently valid U.S. National Standards.

This document has been prepared as part of the service of ISA–The Instrumentation, Systems, and Automation Society, toward a goal of uniformity in the field of instrumentation. To be of real value, this document should not be static but should be subject to periodic review. Toward this end, the Society welcomes all comments and criticisms and asks that they be addressed to the Secretary, Standards and Practices Board; ISA; 67 Alexander Drive; P. O. Box 12277; Research Triangle Park, NC 27709; Telephone (919) 549-8411; Fax (919) 549-8288; E-mail: standards@isa.org.

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Foreword

This standard has been prepared by ISA's SP20 standards committee, Instrument Specification Forms. The ISA–SP20 committee welcomes comments and suggestions, and requests that they be addressed to the Chairperson, ISA-SP20, ISA, 67 Alexander Drive, Research Triangle Park, NC 27709 USA.

This standard modifies the form style of the previous ISA-20-1981 to provide a dedicated page for operating parameters, and expands the form content beyond the limitations of ISA-20-1981.

This standard applies to all processes of development and use of specification forms for process measurement and control instruments. Specifically, it provides requirements and guidance for the development of new or revised forms.

The other parts of this series, *Specification Forms for Process Measurement and Control Instruments,* are as follows:

ISA–TR20.00.01–2001	Part 1: General Considerations
ISA Draft 20.00.02	Part 2: Instructions for Using Specification Forms
ISA Draft TR20.00.04	Part 4: Data Dictionary

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Introduction

Specification forms developed by ISA's SP20 committee are designed to facilitate communication between instrument specifiers and manufacturers in the exchange of data on standard commercially available instrument devices. They are not intended to be used as a hardware design specification. New devices and options for existing devices continually enter the market and may justify the creation or modification of specification forms.

The following are the purposes of these specification form requirements and development guidelines:

a) To minimize the effort and duration required to develop and review new forms for ISA approval

b) To promote uniformity within a form and between forms in content, format, terminology, and maximum field prompt lengths

c) To promote uniformity between form parts by identifying content criteria for operating parameters and device properties

d) To assist in the preparation of a complete specification form by identifying device properties that correlate with the operating parameters

e) To provide guidance to develop instrument specification forms, field prompt contextual help instructions, and pick lists

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1 Scope

This standard establishes requirements for instrument specification forms to be published by ISA. Its use is recommended for organizations that develop their own equivalent specification forms.

The guidelines set forth in this standard will assist in the efficient production of specification forms that maintain consistency and compatibility with ISA specification forms.

2 Normative references

No normative references are required in using this standard. For a list of informative references, see annex A.

3 Definitions

For the purposes of this standard, the following definitions apply.

3.1 contextual help:
explanation of the field prompt or desired data entry field response.

3.2 data entry field:
the specification form area into which data are entered.

3.3 field prompt:
the descriptive text on the specification form that identifies the data entry field.

3.4 main device:
the fundamental or most important measurement or control device of a composite assembly that contains secondary devices or accessories.

3.5 pick list:
a list of example or preferred enumerated choices for a data entry field.

4 Application

Members of standards committees, users or manufacturers, and experts can use their knowledge of particular instrument hardware and this standard to develop new specification forms, provide contextual help instructions, and develop pick lists for filling out the forms.

These form requirements and development guidelines may be used for the following:

a) To evaluate the adequacy of an existing *Operating Parameters* specification form for specifying a new hardware device (as per ISA–TR20.00.01–2001)

b) To develop new instrument *Device Specification* forms (as per ISA–TR20.00.01–2001)

c) To establish review criteria for new or revised forms seeking approval

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5 Requirements and optional information

5.1 Form types, content, and order

5.1.1 Three form types

The specification forms are generally comprised of three form types for the following information organization:

a) *Operating Parameters* forms are developed to communicate the process variables, operating conditions, process design conditions, component design criteria, material properties, and operating environment conditions of a device's application.

b) *Device Specification* forms are developed to communicate hardware device requirements and properties for a main device and for attached and directly associated devices.

c) The *General or Special Requirements* form has been developed to communicate the Table of Contents for general or special requirements that are applicable to the device, but that need additional space beyond that available in the other two types of forms.

5.1.2 Content of *Operating Parameters* forms

Operating Parameters forms are developed to identify the process variables to be measured or controlled, and the related process and environmental parameters that affect the measurement or control objectives. A single form is appropriate for all instrument device types that relate to the identified common measurement variable or final control element.

The following information categories are required for these forms, if appropriate for the device type:

a) Administrative identifications

b) Service identifications

c) Component design criteria

d) Material flow condition variables that are necessary to perform sizing calculations or device type selection

e) Process design conditions that are necessary for device type selection

f) Material name, composition, and properties required for calculations, device sizing, or suitability evaluation

g) Environmental operating conditions

h) Physical installation data that are required to evaluate the device application.

Optional parameters that should be evaluated for inclusion on an *Operating Parameters* form, or on a supplemental page of the form, include the following:

a) Special requirements for the application or hardware device

b) Preliminary sizing analysis calculation results that are pertinent to sizing or device type selection

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c) Basic engineering design data that identify the environmental and service parameters that the device must operate within

d) Detailed material composition data

5.1.3 Content of *Device Specification* forms

Device Specification forms are developed to encompass the information required that defines the physical instrument device and the requirements for its immediately associated secondary devices. The secondary device properties may be developed on a supplemental form page.

The specific content of any specification form should primarily be dependent only upon the characteristics of the device that are necessary for communication between the specifier and the manufacturer. The intended use of the specification form, such as a document for preliminary inquiry and quotation, a traditional specification, or a conforming specification, can affect how much of the form content is required to be used to accomplish the respective objectives.

The general criteria for including a property on a device specification form include the following.

5.1.3.1 Type and style field prompts

Each subsection of the form should start with a “type” field intended to clarify the major classification of that device component. A “style” field is sometimes additionally advantageous to classify component construction.

5.1.3.2 Compliance verification field prompts

Correlated parameter field prompts are required that provide confirmation that the device properties comply with the operating parameter limits that are identified in the specification. Typical field prompt categories that must be considered for inclusion are shown in table 1.

Table 1 — Correlated parameters

Operating Parameters form field category	Device Specification form field category
Component design criteria	Performance characteristics
Process design conditions	Design ratings
Measured variable range	Measurement range limits
Calculated operating variables	Working limits
Piping termination properties	Termination properties
Piping materials	Materials of construction

5.1.3.3 Model designation field prompts

Device manufacturers’ literature generally states their required data for ordering. Field prompts are required to cover those properties by explicitly identifying each of the device attributes that are so identified or used to generate the model number. These field prompts will generally include the following types of properties:

a) Model series that identifies the device type or style

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b) Nominal size

c) Process connection nominal size, category, type, and rating

d) Measurement span and range limits

e) Calibration or adjustment values

f) Control action

g) Signal power source

h) Output signal type and range limits

i) Standard design optional features

j) Material of construction for pressure boundary components

k) Major material of construction for internal wetted components

l) Mounting type or configuration

m) Enclosure classification

n) Standard optional accessories

o) Safety certifications, listings, or approvals

5.1.3.4 User-definable fields

Blank field prompts shall be provided in each form subsection to accommodate specifying of the less common device properties.

5.1.3.5 Optional performance characteristics field prompts

Design and device selection reviews often require some device performance characteristics to be provided by the manufacturer in addition to those covered in table 1.

5.1.3.6 Optional construction design alternatives field prompts

Manufacturers are often given responsibility to select device construction alternatives that should ultimately be reviewed for acceptability by the specifier. Those properties that are generally desirable to solicit and document within the specification form include the following:

a) Enclosure ratings

b) Standard material of construction for pressure boundary components not covered in the basic model number

c) Standard material of construction for wetted components not covered in the basic model number

d) Standard accessories recommended for the service application

e) Optional construction styles or features recommended for the service application

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5.1.3.7 Optional special requirements field prompts

Properties for frequently specified special requirements are desirable, such as the following:

a) Tag or nameplates

b) Preparations

c) Construction code or standard

d) Calibration or inspection reports

5.1.3.8 Optional physical data field prompts

Design and device selection reviews often require some basic device physical data that are advantageous to incorporate in the specification form data, such as the following:

a) Estimated weight

b) Major dimensions

c) Manufacturers' dimensional reference drawing

5.1.3.9 Optional calibrations and tests data field prompts

Documentation of device properties that are used during a maintenance check or replacement is desirable, such as the following:

a) Bench set limits

b) Rated travel

c) Test conditions

5.1.4 Subsection order of *Device Specification* forms

The properties of instrument devices, and their associated secondary devices, have been data modeled to facilitate consistency of presentation between the properties of devices and between specification forms. The specification form should select the appropriate generic property groupings, and arrange the respective specific subsections in the following recommended order:

a) Pressure-containing shell properties

b) Pressure-containing shell extension component properties

c) Internal subcomponent properties

d) Signal input/output extension component properties

e) Integral input/output component properties

f) Auxiliary devices properties

g) Performance characteristic properties

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h) Accessories properties

i) Special requirements

j) Physical data properties

k) Calibrations and tests properties

l) Component identifications

5.1.5 Subsection content order of *Device Specification* forms

The following field prompt order shall be used (as applicable) within any subsection.

5.1.5.1 Assembly or component

a) Type

b) Style

c) Nominal size

5.1.5.2 Process or end connections

a) Nominal size

b) Nominal rating

c) Termination category and type

d) Finish

e) Length and wall thickness

f) Location

g) Taps

5.1.5.3 Part

a) Type

b) Style

c) Quantity

d) Dimensions

5.1.5.4 Materials of construction

a) Primary material

b) Secondary materials

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c) Bolting materials

d) External treatment material

e) End termination material

f) Gasket, packing, and seal materials

g) Accessories material

5.1.5.5 Performance characteristics

a) Design ratings

b) Working limits

c) Accuracy ratings

d) Measuring range limits

e) Performance factors

f) Rated coefficients

g) Performance ratings

h) Environmental limits

i) Supply limits and consumption

j) Signal and communication limits

5.1.5.6 Accessory hardware

a) Style

b) Nominal size

c) Standard options

d) Material of construction

e) Model number

5.1.5.7 Special requirements

a) Custom tag or nameplate

b) Referenced specification identification

c) Special preparation

d) Design compliance standard

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e) Construction code identification

f) Calculation requirement

g) Calibration report

h) Software configuration

i) Certification or approvals documentation

j) Special inspection

5.1.5.8 Physical data

a) Weight

b) Dimensions

c) Signal connection

d) Manufacturer's dimensional reference drawing

5.1.5.9 Calibrations and tests

a) Measurement range values

b) Control action

c) Signal range limits

d) Setpoint values

e) Test values

5.1.5.10 Component identifications

a) Component tag number

b) Component type

c) Style

d) Manufacturer

e) Composite model number

5.1.6 Content of *General or Special Requirements* form

The *General or Special Requirements* form, which is optional, has been developed to provide a form for attached requirements that cannot be adequately described in the *Operating Parameters or Device Specification* forms. The first sheet of this form shall contain a note referencing the requirements as part of the specification, and a Table of Contents of the general or special requirements. The same specification form identification header, revision chronicle, and remarks footer subsections used on the *Operating*

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Parameters or Device Specification forms shall be included. Subsequent pages may use the same or a different format as appropriate for the attached references.

5.2 Form formats

5.2.1 Two form formats

ISA specification forms are provided in one of two format types: single main device format or multiple similar main devices formats. The quantity of the required operating parameters information will generally determine the format type that is appropriate for a specific device specification.

5.2.1.1 Single main device format

There shall be only one main device per specification form. The form may include attached and directly associated secondary devices.

5.2.1.2 Multiple similar main device format

Similar main devices can share a common hardware specification form and utilize a separate tabulation sheet to identify the variable data and unique tags.

5.2.2 Printing format

The page format has been established to readily accommodate typical printer capabilities and minimize the pages required to accommodate the identified required content. The printing style shown on the ISA example specification forms has been chosen for convenience of determining the maximum available prompt and data field lengths. Alternate printing styles may be used if they produce the same results as the following listed page format.

5.2.2.1 Page size

The page size shall be 8½ inches by 11 inches, or A4 size.

5.2.2.2 Columnar presentation

The form shall use multiple parallel columns of field prompts and data entry fields, with the following field or text area sizes (identified in characters), and the desired text alignments within the fields:

Size	Text Area Description	Text Alignment
3	Column for line number	Right
2	Column for revision identification field	Left
25	Column for field prompt area	Left
25 (maximum)*	Data entry field(s)	Left
1	Space between a data entry field and its unit symbol field	
1 (minimum)	Space between a data entry field and a subsequent field prompt	
* The data entry fields are generally limited to 25 total characters, but selected exceptions have been made for special conditions that require more characters.		

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5.2.2.3 Data grouping

Related data shall be grouped as defined in this standard, and a text line shall contain the title for the group.

5.2.2.4 Top margin and print

The top margin shall be a minimum of one line or the printer's minimum capability. The maximum required lines of printing shall be 84.

5.2.2.5 Left margin

The left margin shall be a minimum of 0.6 inch to allow for hole punches.

5.2.3 Numbering system format

5.2.3.1 Page size

There shall be unique line numbers for each form part. These unique numbers serve as references to specific field prompts and data entry fields.

5.2.3.2 Consecutive line numbers within a part

The line numbers that correspond to the field prompts within a subgroup shall be consecutive.

5.3 Form terminology

The same field prompt terminology shall be used throughout all forms, sections, or subsections of a form to designate a given property. Use of an alternate term (synonym) for a concept already defined shall be avoided.

Field prompts from the ISA Draft TR20.00.04 (see annex A) master list shall be used whenever possible.

If new field prompt terms are necessary, they shall be listed and submitted for approval by ISA for inclusion in the ISA Draft TR20.00.04 master listing.

5.4 Data field content

Field prompts should be constructed to require only a single parameter for the data entry field response. Compound parameter responses would generally require complex help instructions, and a large pick list to cover all possible combinations of the individual properties.

Unit symbols relating to numeric data should be placed in separate fields to facilitate enumerated value pick list and data conversion to alternate units.

6 Preparation of new specification forms

The specification forms are developed to reflect the data required for the selection and definition of standard, commercially available instrument devices. The methodology for developing the forms encompasses the application of this standard's requirements while examining reference sources and the guidelines presented in this section.

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6.1 Reference sources

6.1.1 Specification forms

The collection, whenever they are available, of similar specification forms from the following sources will facilitate the development of new forms:

a) ISA forms for similar instruments developed in accord with this standard

b) ISA forms from ISA-20-1981 or other ISA standards

c) Standards organizations such as the International Electrotechnical Commission, Fluid Controls Institute, American Petroleum Institute, American Society of Mechanical Engineers, and Institute of Electrical and Electronics Engineers

d) Instrument manufacturers and vendors

e) Other form users such as those from engineering and construction firms or end-user engineering departments

6.1.2 Reference documents

The gathering, whenever they are available, of the following types of applicable terminology-related reference documents may facilitate the development of specification forms:

a) ISA device-associated standards, recommended practices, and technical reports; and the ISA *Dictionary of Measurement and Control*

b) Related standards from organizations such as the International Electrotechnical Commission, Fluid Controls Institute, American Petroleum Institute, American Society of Mechanical Engineers, and Institute of Electrical and Electronics Engineers

c) Manufacturers' device specifications and associated technical literature

d) Technical publications that contain sizing calculation methods or selection criteria

6.2 Review and modification of *Operating Parameters* forms

The appropriate measured variable or final device *Operating Parameters* form will generally be adequate for new or revised devices, but must be checked against the requirements in 5.1.2. Identify and incorporate any necessary revisions that must be made to the existing *Operating Parameters* form.

6.3 Development guidelines for *Device Specification* forms

6.3.1 Specification identification header and footer

The *Operating Parameters and Device Specification* forms have been developed with fixed specification identification headers and footers. These shall be duplicated for new forms.

6.3.2 Component identification subsection

Device Specification forms have been developed with fixed component identifications subsections. This shall be duplicated for new forms.

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6.3.3 Main device subsections

The main measurement or control device attributes generally can be subdivided into subsections. Determine if such subsection groupings exist for the device under consideration and establish the corresponding device-specific titles for those subsections.

6.3.4 Main device subsection development and review

The form's format can be most easily implemented by editing an existing form for a similar instrument.

Examine numerous manufacturers' products literature and identify their required device property data for specifying the device or component, while encompassing the following:

a) This standard's content requirements identified in 5.1.3

b) The field prompt order identified in 5.1.5

c) The terminology from ISA Draft TR20.00.04 (see annex A) for terms not present on the example form being edited

There will generally be more field prompt candidates, based on the above criteria, than can be accommodated on the form. The criteria may need to be prioritized and compromised to fit page limitations. The user-definable field prompts required in each subsection and the notes subsections will generally provide the flexibility to accommodate most applications.

6.3.5 Secondary device inclusion

The decision of when to include in the form the specification of an attached or directly associated device should consider the following:

a) Forms should provide for attached equipment that is commonly used and available as a standard option by most manufacturers

b) A supplemental page can be developed to provide for less commonly used attached or directly associated devices, such as has been provided for the control valves specification form

c) If it is desirable to have a second supplier provide the directly associated device, a separate form for the device should be produced

6.4 Contextual help instructions

The contextual help instructions should clarify the field prompt's context or definition and, preferably, be applicable to all devices using that field prompt. Therefore, the only new contextual help instructions that should need development will occur when new field prompts are added to ISA Draft TR20.00.04 (see annex A).

6.5 Pick list

Review of manufacturers' literature and ISA Draft TR20.00.04 (see annex A) should be made to determine the adequacy of the existing list to cover the new device. Unit symbols should be provided in both US Customary and SI systems, and equivalent dimensions provided when available. New prompts should be based on terminology from reference documents such as those identified in 6.1.2.

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6.6 Data dictionary update

If ISA Draft TR20.0.04 (see annex A) does not contain adequate terms for the device specification under development, any proposed new data dictionary terms shall be submitted for approval along with the new or revised form.

6.7 Review and approval

After completing the form development with this standard's requirements and guidelines, circulate the preliminary form for review by peers and manufacturers. Consolidate the comments and send the proposed form to the Chairperson of ISA-SP20 as a request for interpretation of standards, for the proposed new or revised form, as an addendum or future revision to ISA–TR20.00.01–2001.

7 Classification and designation

Hardware device forms are grouped and numbered based upon the device's measurement principles or control functions. Consult ISA–TR20.00.01–2001, table A.1 or table A.2, when tentatively assigning new device categories, form numbers, and titles.

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Annex A (informative) — References

This annex is included for informational purposes and is not part of this standard. At the time of publication, the editions or drafts indicated below were valid. All standards, recommended practices, and technical reports are subject to revision, and parties to agreements based on this standard are encouraged to investigate the possibility of applying the most recent editions or drafts of the documents indicated below.

ISA

ISA–TR20.00.01–2001	Specification Forms for Process Measurement and Control Instruments Part 1: General Considerations
ISA Draft 20.00.02	Specification Forms for Process Measurement and Control Instruments Part 2: Instructions for Using Specification Forms
ISA Draft TR20.00.04	Specification Forms for Process Measurement and Control Instruments Part 4: Data Dictionary
Available from:	ISA–The Instrumentation, Systems, and Automation Society 67 Alexander Drive PO Box 12277 Research Triangle Park, NC 27709 USA Tel: (919) 990-9200 www.isa.org

ANSI

ANSI X3.50-1986 (R1997)	Representations for U.S. Customary, SI, and Other Units to Be Used in Systems with Limited Character Sets
Available from:	American National Standards Institute 11 West 42nd Street New York, NY 10036 USA Tel: (212) 642-4900 www.ansi.org

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ASME

ASME Y1.1-1989	Abbreviations for Use on Drawings and in Text
Available from:	ASME 3 Park Avenue New York, NY 10016-5990 USA Tel: (973) 882-1167 www.asme.org

International Electrotechnical Commission (IEC)

IEC 60050-351 (1998-09)	International Electrotechnical Vocabulary - Part 351: Automatic Control
Available from:	IEC 3, rue de Varembe PO Box 131 1211 Geneva 20 Switzerland Tel: 41 22 734 0150 www.iec.ch

International Organization for Standardization (ISO)

ISO/CD 10303-221	Library of Classes for Process Plant Application Protocols
Available from:	ISO 1, rue de Varembe PO Box 56 1211 Geneva 20 Switzerland Tel: 41 22 749 0111 www.iso.ch

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